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A comparison of fisheries biological reference points estimated from temperature-specific multi-species and single-species climate-enhanced stock assessment models.

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ABSTRACT

Multi-species statistical catch at age models (MSCAA) can quantify interacting effects of climate and fisheries harvest on species populations, and evaluate management trade-offs for fisheries that target several species in a food web. We modified an existing MSCAA model to include temperature-specific growth and predation rates and applied the modified model to three fish species, walleye pollock (*Gadus chalcogrammus)*, Pacific cod (*Gadus macrocephalus*) and arrowtooth flounder (*Atheresthes stomias*), from the eastern Bering Sea (USA). We fit the model to data from 1979 through 2012, with and without trophic interactions and temperature effects, and use projections to derive single- and multi-species biological reference points (BRP and MBRP, respectively) for fisheries management. The multi-species model achieved a higher over-all goodness of fit to the data (i.e. lower negative log-likelihood) for pollock and Pacific cod. Variability from water temperature typically resulted in 5-15% changes in spawning, survey, and total biomasses, but did not strongly impact recruitment estimates or mortality. Despite this, inclusion of temperature in projections did have a strong effect on BRPs, including recommended yield, which were higher in single-species models for Pacific cod and arrowtooth flounder that included temperature compared to the same models without temperature effects. While the temperature-driven multi-species model resulted in higher yield MBPRs for arrowtooth flounder than the same model without temperature, we did not observe the same patterns in multi-species models for pollock and Pacific cod, where variability between harvest scenarios and predation greatly exceeded temperature-driven variability in yield MBRPs. Annual predation on juvenile pollock (primarily cannibalism) in the multi-species model was 2-5 times the annual harvest of adult fish in the system,

thus predators represent a strong control on population dynamics that exceeds temperature-driven changes to growth and is attenuated through harvest-driven reductions in predator populations. Additionally, although we observed differences in spawning biomasses at the accepted biological catch (ABC) proxy between harvest scenarios and single- and multi-species models, discrepancies in spawning stock biomass estimates did not translate to large differences in yield. We found that multi-species models produced higher estimates of combined yield for aggregate maximum sustainable yield (MSY) targets than single species models, but were more conservative than single-species models when individual MSY targets were used, with the exception of scenarios where minimum biomass thresholds were imposed. Collectively our results suggest that climate and trophic drivers can interact to affect MBRPs, but for prey species with high predation rates, trophic- and management-driven changes may exceed direct effects of temperature on growth and predation. Additionally, MBRPs are not inherently more conservative than single-species BRPs. This framework provides a basis for the application of MSCAA models for tactical ecosystem-based fisheries management decisions under changing climate conditions.

"Behind these numbers lies, of course, an infinity of movements and of destinies."

– von Bertalanffy 1938

1. Introduction

Increasing global demand for food resources (Delgado et al., 2003) concomitant with climate-driven alterations to marine food webs and declining trends in fisheries catch per unit effort worldwide, suggests little room for fisheries expansion and increased need for efficient and holistic fisheries management approaches (Merino, 2012; Worm and Branch, 2012; Link 2010; Link and Browman 2014). Ecosystem-based harvest strategies are pivotal for sustainable fisheries management in a changing and uncertain future (US commission on Ocean Policy, 2004; MEA, 2005; Murawski and Matlock, 2006; Fogarty 2014), and of paramount importance is the ability to distinguish fishery impacts from large-scale climate drivers and short-term physical and trophodynamic processes (Murawski and Matlock, 2006; Hollowed et al., 2009; 2013; Gamble and Link, 2012). Ecosystem-based fisheries management (EBFM) has been proposed as a place-based, rather than species-based, management approach (Fogarty 2014) that can account for fishery effects on food web structure and function, balance biodiversity and yield objectives, and improve management in systems characterized by large seasonal, decadal, and long-term climate variability (Pikitch et al., 2004; Arkema et al., 2006; Link 2010; Agardy et al., 2011; Levin et al., 2013; Link and Browman 2014).

A variety of analytical tools have been developed that can address EBFM needs (see reviews in Hollowed et al., 2000, 2009, 2013; Plagányi et al., 2014, Link 2010, Link and Browman 2014, Fogarty 2014). Multi-species statistical catch-at-age models (MSCAA) are an example of a class of multi-species 'Models with Intermediate Complexity for Ecosystem assessments' (i.e., MICE; Plagányi et al., 2014), which have particular utility in addressing both strategic and tactical EBFM questions (Hollowed et al. 2013; Fogarty 2014; Link and Browman 2014; Plagányi et al., 2014). MSCAA models may increase forecast accuracy, may be used to evaluate propagating effects of observation and process error on biomass estimates (e.g., Curti 2013; Ianelli et al., this issue), and can quantify climate and trophic interactions on species productivity. As such MSCAA models can address long recognized limitations of prevailing single species management, notably non-stationarity in mortality and maximum sustainable yield (MSY), and may help reduce risk of overharvest (Link 2010; Plagányi et al., 2014; Fogarty 2014). Because multispecies biological references points (MBRPs) from MSCAA model are conditioned on the abundance of other species in the model (Collie and Gislason 2001; Plagányi et al., 2014; Fogarty 2014), they may also have utility in setting harvest limits for multi-species fleets, evaluating population dynamics in marine reserves or non-fishing areas, and quantifying trade-offs that emerge among fisheries that impact multiple species in a food web (see reviews in Pikitch et al., 2004; Link 2010; Levin et al., 2013; Link and Browman 2014; Fogarty 2014).

Depending on their structure, MSCAA models can be used to evaluate climate- and fisheries-driven changes to trophodynamic processes, recruitment, and species abundance (Plagányi et al., 2014). MSCAA models differ somewhat among systems and

species, but most use abundance and diet data to estimate fishing mortality, recruitment, stock size, and predation mortality simultaneously for multiple species in a statistical framework. Similar to age structured single species stock assessment models widely used to set harvest limits, MSCAA models are based on a population dynamics model, the parameters of which are estimated using survey and fishery data and maximum likelihood methods (e.g., Jurado-Molina et al., 2005; Kinzey and Punt, 2009; Van Kirk et al., 2010; Kempf 2010; Curti et al., 2013; Tsehaye et al., 2014). Unlike most single-species models, MSCAA models additionally separate natural mortality into residual and annually varying predation mortality, and model the latter as a series of predator-prey functional responses. Thus, natural mortality rates for each species in MSCAA models depend on the abundance of predators in a given year and vary annually with changes in recruitment and harvest of each species in the model.

MSCAA models have specific utility in quantifying direct and indirect effects of fisheries harvest on species abundance and size distributions (see reviews in Hollowed et al., 2000a, 2013; Link 2010; Fogarty 2014; Link and Browman 2014; Plagányi et al., 2014), which is important for EBFM and trade-off analyses of various management strategies. Rapidly shifting climate conditions are also of growing concern in fisheries management as changes in physical processes are known to influence individual growth, survival, and reproductive success of fish and shellfish (Hanson et al., 1997; Kitchell et al., 1977; Morita et al., 2010; Hollowed et al., 2013, Cheung et al., 2015). Climate-driven changes in water temperature can directly impact metabolic costs, prey consumption, and somatic or gonadal tissue growth, with attendant indirect effects on survival, production, and sustainable harvest rates (e.g., Hanson et al., 1997; Morita et al., 2010, Cheung et al.,

2015). Temperature-dependent predation, foraging, metabolic, and growth rates are common in more complex spatially-explicit food web or whole of ecosystem models such as GADGET (e.g., Howell and Bogstad 2010; Taylor et al., 2007), Atlantis (e.g., Fulton et al., 2011; Kaplan et al., 2012; 2013), and FEAST (Ortiz et al., this issue). Temperature functions for growth and predation can also be incorporated into MSCAA models, allowing this class of models to be used to evaluate interacting climate, trophodynamic, and fishery influences on recommended fishing mortality rates.

Numerous studies point to the importance of using multi-species models for EBFM (see review in Link 2010). Multi-species production models produced different estimates of abundances and harvest rates than single species models for Northeast US marine ecosystems (Gamble and Link, 2009; Tyrrell et al., 2011), and MSY of commercial groundfish stocks estimated from aggregated production models are different than the sum of MSY estimates from single-species assessments (Mueter and Megrey, 2006; Gaichas et al., 2012; Smith et al., 2015). Multi-species models have been used to demonstrate long-term increases in yield of Icelandic stocks of Atlantic cod (*Gadus morhua*) and reductions in capelin (*Mallotus villosus*) and Northern shrimp (*Pandalus borealis*) catch associated with short-term decreases in cod harvest (Danielsson et al., 1997). Kaplan et al. (2013) demonstrated the disproportionately large ecosystem impacts of applying the same Fx% (e.g., Fx%, or the harvest rate that reduces spawning stock biomass to x% of unfished spawning stock biomass, SSB_0; Caddy and Mahon, 1995; Collie and Gislason, 2001) harvest control rule approach to forage fish as is used for groundfish in the northeast Pacific, and trophodynamics in a southern Benguela

ecosystem resulted in higher carrying capacity for small pelagic species under fishing (versus no-fishing) scenarios (Smith et al., 2015).

Since natural mortality and recruitment rates in a MSCAA model are conditioned on harvest rates of predators in the model, an ongoing area of research is evaluating MSCAA model analogs to single-species biological reference points (see Moffitt et al., this issue), such as harvest rates that correspond to maximum yield (F_{MSY}) or proxies thereof (e.g., Fx%). Other multi-species models have been used to derive and evaluate MBRPs, although these have largely focused on MSY (e.g., Kaplan et al., 2013; Smith et al., 2015). A notable exception is Collie and Gislason (2001), who evaluated a variety of MBRPs using a multi-species, virtual population analysis and found MBRPs to be sensitive to variation in natural mortality (much less so to variability in growth), and as such proposed that fishing mortality reference levels for prey species with high mortality be conditioned on the level of predation mortality. Building on this approach, Moffitt et al. (this issue) recently demonstrated a projection approach for using multi-species models to derive a variety of MBRPs for EBFM. This provides a basis for the application of MSCAA models for increased use in tactical and strategic EBFM decision-making across a diversity of management frameworks worldwide.

Operationalizing regional MSCAA models is of particular interest in Alaska (USA), where regional fisheries management is progressively moving towards EBFM. The region, especially the eastern Bering Sea, is defined by large, climate-driven changes to trophodynamics and species productivity that can vary on annual and multi-annual timescales (see reviews in Aydin and Mueter 2007; Stabeno et al., 2012; Baker et al., 2014). Accordingly, fisheries management in Alaska has a long history of using

ecosystem information and multi-species models for strategic management advice (e.g., multi-species model-based indices, such as mean trophic level, are regularly reported in the annual Ecosystem Considerations chapter of Alaska Stock Assessment and Fishery Evaluation (SAFE) reports; see review in Livingston et al., 2011). Development of multiple MSCAA models in the region (Jurado-Molina et al., 2005; Kinzey and Punt , 2009; Van Kirk, 2010) has advanced regional management closer to EBFM, facilitating use of estimates from MSCAA models in single-species models used for tactical decisions in the region. For instance, Dorn et al. (2014) recently used predation mortality estimates from a regional MSCAA model developed by Van Kirk (2010) to inform natural mortality for the Gulf of Alaska walleye pollock (*Gadus chalcogrammus*, hereafter "pollock") stock assessment.

MSCAA models may be most useful for species that exhibit strong trophic interactions (predator and prey species) or contrasting management or biological constraints that require simultaneous evaluation (Link 2010). In the eastern Bering Sea, pollock support one of the largest fisheries worldwide, with over 1.2 million metric tons (t) harvested per year (representing ~99% of the annual quota; Ianelli et al., 2014). Pollock are both predators (adults) and prey (i.e., ages <2; Dunn and Matarese, 1987; Nishiyama et al., 1986) for a variety of species including cannibalistic conspecifics (e.g., Boldt et al., 2012). Variable climate conditions, particularly the spatial extent of winter sea ice, the timing of sea ice spring melt, and subsequent summer bottom temperatures, can differentially promote survival of pollock and their predators and/or modulate predator and prey overlap in the region (e.g., Baily 1989; Zador et al., 2011; Boldt et. al 2012; Baker and Hollowed 2014). Diet analyses suggest Pacific cod (*Gadus*

macrocephalus), cannibalistic conspecifics, and arrowtooth flounder (*Atheresthes stomias*), amongst others, are important predators of pollock populations in the eatern Bering Sea (Livingston 1993; Aydin and Mueter 2007). Arrowtooth flounder have been increasing in recent years, possibly in response to climate conditions that favor their recruitment (e.g., Zador et al., 2011; Hunsicker et al., 2013), yet a combination of low flesh marketability and high bycatch rates of valuable flatfish species limits their annual harvest. This provides an interesting contrast to highly valuable pollock and Pacific cod fisheries, which favor cold conditions projected to decline under future climate scenarios (Mueter et al., 2011; Ianelli et al. 2011; Stabeno et al., 2012).

Here we present a three species MSCAA model for the Bering Sea (hereafter CEATTLE, for Climate-Enhanced, Age-based model with Temperature-specific Trophic Linkages and Energetics) that includes temperature-dependent von Bertalanffy weight-at-age functions (VBGF; von Bertalanffy, 1938) and temperature-specific, bioenergetics-based predation interactions. Using U.S. National Marine Fisheries Service Alaska Fisheries Science Center (AFSC) survey and fishery data, we estimate retrospective temperature- and size-specific predator rations, prey preference, and weight-at-age to evaluate how predation mortality, recruitment, and survival of three target species change under historical climate conditions and harvest rates. We then project CEATTLE forward to derive estimates of unfished biomass and fishing mortality under future climate conditions and various harvest scenarios (*sensu* Curti et al., 2013; Moffitt et al., this issue; Ianelli et al., this issue) to evaluate the sensitivity of proxy single- and multi-species biological reference points (BRP and MBRP, respectively) to harvest scenarios and trophic interactions.

2. Methods

2.1 Model structure

Programmed in AD model builder (Fournier et al., 2012), CEATTLE is an example of an "environmentally-enhanced" stock assessment model (*sensu* Link 2010), where temperature-specific algorithms predict size-at-age and predation mortality. CEATTLE builds on earlier models that combine catch-at-age assessment models with multi-species virtual population analysis (MSVPA) in a statistical framework (Jurado-Molina et al., 2005). Abundance and biomass of each cohort is modeled using standard population dynamics equations, accounting for a plus age group (Table 1, Eqs. T1.3, T1.4, T1.5). The initial age-structure is assumed to correspond to unfished equilibrium, and the numbers of each species at age 1 each year are treated as estimable parameters (Eqs. T1.1 and T1.2). Total mortality of each prey species *i* age *j* (or predator species *p* age a) in each year *y* is the sum of mortality due to predators in the model ($M2_{ij,y}$), fishing mortality ($F_{ij,y}$), and residual mortality ($M1_{ij}$), Eq. T.1.8). Predation mortality (Eq. T2.1) is based on the assumption that the annual age-specific ration of a predator is allocated to prey species of a given age according to predator selectivity (Table 2, Eq. T.2.2). Predator selectivity is based on the suitability function derived by Jurado-Molina et al. (2005) and fit to available data from 1981-2012, while annual ration is a function of temperature-specific allometric relationships between ration and fish weight based on bioenergetics models for each species (Eqs. T2.4 and T2.5; S12; see Holsman and Aydin, 2015 for more detail).

The length-to-weight relationships, predator size and species diet preference,

bioenergetics-based, temperature-specific predator rations, and maturity were based on previous studies (Tables 1 and 2; S10, S11, S12; Holsman and Aydin, 2015). Size-specific diet compositions for each species were assumed known based on diet data collected during the AFSC bottom trawl survey (i.e., diet data are not included in the objective function) and trophic patterns in survey and fishery-based diet data were used to calculate mean (across years and stations) predator-prey suitability (Eq. T2.2).

Water temperature is known to directly impact growth through influencing metabolic and digestion rates, which often scale exponentially with body weight and temperature (see Hanson et al., 1997 for an overview). Thus we modified the generalized formulation of the Von Bertalanffy growth function (VBGF; von Bertalanffy 1938; Pauly 1981; Temming 1994) to predict temperature-dependent growth by allowing the allometric scaling parameter *d* to increase with temperature. Essington et al. (2010) and Holsman and Aydin (2015) describe the derivation and application of the VBGF towards bioenergetics modeling in great detail, so we do not repeat it here. Essentially, in this formulation *d* represents the realized allometric slope of consumption, which integrates both the direct effect of temperature on consumption and indirect ecological interactions that scale with temperature and influence relative foraging rates (see Essington et al., 2010; Holsman and Aydin, 2015). We fit the VBGF to otolith-based length- and weight-at-age data (n = 21388, 14362, and 772, for pollock, Pacific cod, and arrowtooth flounder, respectively) collected during AFSC Bering Sea surveys and analyzed at the AFSC such that:

$$W_{ij,y} = W_{\infty,iy}\left(1 - e^{\left(-K_i(1-d_{i,y})(j-t_{0,i})\right)}\right)^{\frac{1}{1-d_{i,y}}} e^{\varepsilon}, \text{ where } \varepsilon \sim \mathrm{N}\left(0, \sigma^2_{d,i}\right) \qquad \text{Eq. 1}$$

where $t_{0,i}$ is the age at which $W_{ij,y} = 0$, $W_{\infty,iy}$ is the asymptotic mass which can vary by species *i* and year *y* (i.e., $W_{\infty,iy} = \left(\frac{H_i}{K_i}\right)^{1/(1-d_{i,y})}$), H_i is the assimilation constant K_i is the energy loss constant (Essington et al., 2010), and where ε is a normally and independently distributed random variable with mean 0 and variance $\sigma^2_{d,i}$. Essington et al. (2010) and Holsman and Aydin, (2015) statistically estimated the *d, K* and *H* parameters for various species to estimate consumption rates. In particular, Holsman and Aydin (2015) found that the *d* parameter varied between species and regions in Alaska (USA). We further modified this approach to estimate *d* annually for each year *y* in the dataset, as a linear function of temperature T_y such that:

$$d_{i,y} = e^{(\alpha_{d,i,y} + \alpha 0_{d,i} + \beta_{d,i} T_y)} \quad \text{Eq. 2}$$

where $\alpha 0_{d,i}$ and $\alpha_{d,i,y}$ represent the mean *d* intercept and $\beta_{d,i}$ is the coefficient for the residual effect of temperature on the *d* consumption parameter. We chose this formulation based on the empirical relationship between temperature and consumption, assuming that *d* would capture the differential effects of temperature on growth, and that waste rates scale proportionally with weight but do not vary over time with diet or temperature (i.e. *K* is constant but *d* can vary with temperature). This formulation allows both the slope and asymptotic limit of growth to vary with temperature. Similar approaches, with slightly different modifications to the VBGF, including temperature and prey specific terms for *d* and *k*, respectively, have been used elsewhere to evaluate climate impacts on fish growth (e.g., Cheung et al., 2015; Hamre, 2003).

We used this approach to derive annual temperature-specific coefficients of *d* for pollock and Pacific cod (combined sexes) and separately for male and female arrowtooth

flounder (Table 3; S10). For arrowtooth flounder, we then used the age-specific proportions of mature females (ρ_{ij}) and males ($1-\rho_{ij}$) to derive the mean weight-at-age for both sexes combined (Eq. T1.19 and S6, S7, and S10). Lastly, male and female natural mortality rates (M_{male} and M_{fem}, respectively) and age-specific maturity proportions (ϕ_{ij}) from the 2012 stock assessments for eastern Bering Sea pollock (Ianelli et al., 2012), and Bering Sea and Aleutian Islands Pacific cod (Thompson and Lauth, 2012) and arrowtooth flounder (Spies et al., 2012), were used to derive estimates of the proportion of mature females at age (ρ_{ij}; Eq. T1.18, S7).

2.2. Parameter estimation

The parameters of the model are either pre-specified or estimated by selecting parameters that minimize the log-likelihood function (Table 3) and include fishing mortality rates ($F_{ij,y}$), fishery and survey selectivity (s_{ij}^{f} and s_{ij}^{s}, respectively), initial (pre-harvest) abundance in year 1979 ($N_{0,ij}$), and annual recruitment ($R_{i,y}$), while the estimable parameter of the likelihood function is the catchability coefficient for the acoustic survey (q_1^{eit}; Table 3; Table 4; Fig. 1). We fit the model to available survey and fishery data for the eastern Bering Sea including biomass estimates and age-composition data from the annual AFSC summer bottom trawl survey (Eqs. T4.1 and T4.2), biomass and age-composition data from the AFSC EIT survey (pollock only) (Eqs. T4.3 and T4.4), and the total fishery catch and fishery age-composition data collected by AFSC observers and analyzed at AFSC (Eqs. T4.5 and T4.6; Hilborn and Walters, 1992; Quinn and Deriso, 1999). Penalties were imposed on the changes over age in fishery selectivity (Eq. T4.7). Likelihood priors were applied to normalize the log of annual recruitment and

the fisheries mortality deviations, as well as initial abundances (Eqs. T4.8-T4.10). Selectivity for the AFSC echo integration-trawl (EIT) survey was set to previously reported values (Table 3; Honkalehto et al., 2011; Ianelli et al., 2012).

To evaluate the sensitivity of the model to the temperature-specific predation and growth rates, we also contrasted the single- and multi-species models fit to observed bottom temperatures to the same models where temperature was held constant at mean historical values of 2.51 ^{o}C in the estimation and projection phases. Contrasting the relative change in the single-species model estimates with and without a temperature effect (i.e., with a constant hindcast) can provide insight about the relative effect of the temperature-specific VBGF on the model whereas comparisons between multispecies models can reveal sensitivity of both the VBGF and predation functions to variation in temperature.

2.3. Harvest scenarios and reference points

For all future scenarios, we used an ensemble mean of downscaled Intergovernmental Panel on Climate Change (IPCC AR3) projections for eastern Bering Sea sea surface temperatures (SST; see Ianelli et al., 2011; this issue, for methods) and were converted to bottom temperature (T) using predicted values from a generalized additive model (GAM) fit to annual mean SST and bottom temperatures recorded during the AFSC bottom trawl survey (Fig 2).

We used the approach for deriving biological reference points (BRPs) proposed by Moffitt et al. (this issue) and evaluated 12 harvest scenarios. For each harvest scenario x we calculated female spawning stock biomass for species i at a given fishing mortality

rate F ($SSB_{F,x,i,y}$) by projecting the model forward to 2050 (39 years) under mean recruitment and according to a specified harvest rate (F; $F = 0$ for no fishing scenarios). Here we adopted the current over fishing limit (OFL) tier 3 acceptable biological catch ABC and MSY proxies for Bering Sea groundfish stocks; 40% of unfished biomass as the proxy target biomass for the ABC, and 35% as the proxy for B_{MSY} (female spawning biomass corresponding to maximum sustainable yield, MSY, i.e., 35% of $\overline{SSB}_{0,i}$; Punt et al., 2014; NPMFC, 2013; Clark et al., 1991; Brooks et al., 2010).

The corresponding species-specific, acceptable biological catch ($ABC_{x,i,y}$) for each harvest scenario was calculated as the fishery yield for each year *y* of the projection period [1, n_y^{fut}] given a constant fishing mortality rate for the projection period that satisfies each harvest scenario objective ($F^*_{ABC,x,i}$), such that:

$$ABC_{x,i,y} = \left(\sum_j^{A_i}\left(\frac{F^*_{ABC,x,i}s^f_{ij}}{Z_{x,ij,y}}(1-e^{-Z_{x,ij,y}})N_{x,ij,y}W_{ij,y}\right)\right) \qquad \text{Eq. 3}$$

where $Z_{x,ij,y}$ is the control-rule specific total annual mortality for species *i* age *j* in the set [1, 2,… A_i], s^{f}_{ij} is fishery age selectivity, and $N_{x,ij,y}$ and $W_{ij,y}$ are the annual species-specific abundance and weight-at-age for each projection year *y*. Using this approach, the following harvest scenarios were evaluated (Table 5):

2.3.1 (Scenario 1) Biomass proxy for individual ABC

This harvest scenario is closest to current groundfish OFL tier 3 harvest control used for setting ABC and OFL for Alaskan groundfish species (including those in this study) where assessment models are projected forward 1 and 2 years to determine BRPs for harvest recommendations. This harvest scenario involves finding the species-specific

fishing mortality rate ($F^*_{x,i}$) that results in mean female spawning biomass ($\overline{SSB}_{F,i}$) in the target projection period (i.e., last 5 years; 2046-2050) under fishing that is equal to the target proxy percentage (i.e., 40%) of mean unfished female spawning biomass ($\overline{SSB}_{0,i}$; Table 5). To find $F^*_{ABC,x,i}$, we iteratively project the model to find the $\overline{SSB}_{F,i}$ that corresponds to a given harvest rate $F^*_{x,i}$, adjusting $F^*_{x,i}$ downwards if $\overline{SSB}_{F,i}$ is below the target or upwards if $\overline{SSB}_{F,i}$ is above the target, until we achieve $\overline{SSB}_{F,i}$ near or at the proxy of 40% of $\overline{SSB}_{0,i}$. We ran this harvest scenario with the following variations:

1.1. Find the ABC proxy biomass of 40% of unfished spawning biomass, where unfished biomass ($SSB_{0,i}$) is determined from projections where F is set to 0 for all species simultaneously (Eq. T5.1.1)

1.2. Find the ABC proxy biomass of 40% of unfished spawning biomass as in 1.1, but given a minimum biomass constraint where $SSB_{F,i}$ during the simulation period must be greater than 35% of unfished female spawning biomass (i.e., always above the traditional proxy for the biomass corresponding to MSY; Eq. T5.1.2).

1.3. First find the ABC proxy biomass of 40% of unfished biomass for Pacific cod and arrowtooth flounder (i.e., $\overline{SSB}_{0,i}$, where i = 2 and 3, respectively), while there is no fishing mortality on pollock (Eq. T5.1.3a). Then find the ABC proxy biomass of pollock based on an unfished female spawning biomass for pollock ($\overline{SSB}'_{0,1}$) given Pacific cod and arrowtooth populations are fished at the harvest rate corresponding to the ABC proxy for those two species (Eq. T5.1.3b).

1.4. Find the fishing mortality rate for Pacific cod that corresponds to the ABC proxy biomass of 40% of unfished female spawning biomass when there is no

fishing for arrowtooth flounder and pollock (Eq. T5.1.4a). Next, set the unfished female spawning biomass for arrowtooth flounder conditional on this fishing mortality rate ($\overline{SSB}'_{0,3}$), and find the proxy fishing mortality rate for arrowtooth flounder so that $\overline{SSB}_{F,3}$ is 40% of $\overline{SSB}'_{0,3}$ (Eq. T5.1.4b). Finally, set the unfished female spawning biomass for pollock, $\overline{SSB}'_{0,1}$, conditional on the fishing mortality rates for Pacific cod and arrowtooth flounder, and determine the fishing mortality rate for pollock so that $\overline{SSB}_{F,1}$ is 40% of $\overline{SSB}'_{0,1}$ (Eq. T5.1.4c).

2.3.2 (Scenario 2) Spawning biomass-per-recruit (SPR) approach

This harvest scenario (2.1. in Table 5) involves using CEATTLE model-generated weight-at-age, selectivity and natural mortality rates in a spawning biomass-per-recruit (SPR) analysis for each species *i* to determine the fishing mortality rate that yields a spawning potential ratio ($SPR_{F,i}/SPR_{0,i}$) of 0.4 (Eq. T5.2.1a). This approach assumes that the ABC corresponds to a harvest rate that yields a SPR under fishing that is 40% of the ratio of SPR without fishing. The approach uses the mean predation mortality for each species from the estimation years (1979-2012); weights-at-age, selectivity, and natural mortality rates are fixed at those when all species are simultaneously unfished ($SPR_{0,i}$). The corresponding fishing mortality rate is then used to calculate $\overline{ABC}_{x,i}$ using Eq.1 and $\overline{SSB}_{F,i}$ through projecting the model forward to 2050.

2.3.3 (Scenario 3) Biomass proxy for aggregate ABC

This harvest scenario involves finding the fishing mortality rates consistent with a

species-aggregated fished female spawning biomass (i.e. all three species combined) that is 40% of the species-aggregated unfished female spawning biomass. Here individual species can be fished below the target if the aggregate (i.e., summed) biomass remains above the target.

3.1 Similar to 1.1, iteratively find the system-wide fishing mortality rate (F_{ALL}) that yields a total female spawning biomass that is 40% of the total unfished female spawning biomass. The species-specific fishing mortality rate is F_{ALL} scaled by species-specific natural mortality (Eq. T5.3.1). The scaling by natural mortality reflects that species with higher natural mortality tend to be more productive.

3.2 As for 3.1, except that female spawning biomass for each species is not permitted to drop below 35% of the corresponding unfished female spawning biomass during the projection period (Table 5, Eq. T5.3.2).

3.3 As for 3.1, except that fishing mortality is scaled in proportion to species-specific unfished female spawning biomass (Eq. T5.3.3).

3.4 As for 3.3, except that female spawning biomass for each species is not permitted to drop below 35% of corresponding unfished female spawning biomass during the projection period (Eq. T5.3.4).

2.3.4 (Scenario 4) Maximum yield (MSY)

Many single-species assessments solve spawner–recruit or surplus production relationships for MSY assuming populations are at equilibrium. This approach is analytically difficult in multi-species models because the ratio of recruits per spawner for a given species in multi-species model depends on the population size of other predators

or prey in the model. Thus multi-species models are typically solved to find the "true" maximum yield of all species in the model (e.g., Smith et al., 2015). This harvest scenario involves using a simplex optimization algorithm (i.e., Nelder-Mead; Nelder and Mead 1965) to find the harvest rate for all three species that maximizes aggregate yield.

4.1. Iterate (i.e., eight iterations of the optimization algorithm per species) to find species-specific harvest rates ($F_{x,i}$) that maximize each individual species' yields ($\overline{ABC}_i$) during the last 5 years of the simulation (based on fishing mortality rates for the other species from the previous iteration; Eq. T5.4.1).

4.2. Iterate (i.e., eight iterations of the optimization algorithm) to find the species-specific fishing mortality rates that maximize the total combined yield (i.e., sum of yield for all three species) over the last 5 years of the projection period (Eq. T5.4.2).

4.3. As for 4.2, except that female spawning biomass for each species is not permitted to drop below 35% of the corresponding unfished female spawning biomass (Eq. T5.4.3).

3. Results

3.1. Model parameterization

The multi-species CEATTLE model achieved a higher over-all fit to the data (i.e., lower negative log-likelihood with the same number of estimated parameters for both models) for pollock and Pacific cod. We observed a 10-12% increase in R^2 in fits to survey biomass and age composition data from the single-species (i.e., $M2_{ij,y}$ set to 0, hereafter "single-species model") to multi-species modes of CEATTLE (see S1 for more

detail). Although both models predicted similar total and female spawning biomass, inclusion of trophic interactions in the multi-species model resulted in higher and lower estimates of total biomass for pollock and Pacific cod, respectively (Fig. 3).

Temperature-specific VBGF had the largest effect on Pacific cod biomass estimates in the model, with the single species model exhibiting ~20% lower survey biomass estimates than the same model without temperature (i.e., temperature held constant during the hind casts and mean observed values of 2.51 ^{o}C; Fig. 4). Effects on the single species pollock and Pacific cod VBGF were comparably low to VBGF effects on the multispecies models for all three species, where variability from temperature typically resulted in 5-15% changes in spawning, survey, and total biomasses. Temperature variability largely did not impact recruitment estimates or mortality in the model; recruitment in models with temperature only exhibited 0.25-1% differences from those with constant temperature, and predation mortality only varied 1-7% from the multi-species model with constant temperature (Fig. 5).

In contrast, predation mortality varied considerably with changes in predator abundance over time (Fig. 5). Cannibalism was the largest source of predation mortality for pollock (Figs. 6 and 7) with older conspecifics exhibiting a high preference (i.e., total pollock suitability >0.75) for juvenile pollock (ages 1-3; S3.g). Larger pollock also appear to target small arrowtooth flounder, as evidenced by a slight increase in total suitability of arrowtooth for pollock ages 6-10 (S3.g). Similarly, younger Pacific cod (ages 2-6) also target arrowtooth flounder (S3.h). Pacific cod increasingly target pollock prey as they age, and larger, older Pacific cod diets are dominated by age 1 pollock prey. Pacific cod also appear to be cannibalistic from ages 4 through 9. In contrast arrowtooth

flounder prefer pollock throughout their lives, with total suitability coefficients (for all pollock ages) between 0.5 and 1.0 for arrowtooth flounder ages 1 through 18 (S3.i).

Natural mortality ($M1_{ij}$+$M2_{ij,y}$) was generally highest for age 1-3 pollock and age 1 Pacific cod and arrowtooth flounder (S6). Age 1 pollock exhibited the highest predation mortality, which peaked between 1980-2000 (with a peak of 2.32 yr^{-1} in 1987), and has been relatively stable since then, at a rate of around 1.85 yr^{-1} (Fig. 5a). Mortality was lower for Pacific cod and arrowtooth flounder, with total age 1 natural mortality stable at around 0.66 and 0.60 yr^{-1}, respectively, although both were slightly higher in the first 20 years of the modeled period (Fig. 5b,c). High predation mortality estimated for 1980-1993 for pollock reflects patterns in combined annual demand for prey by all three predators that was highest in the mid 1980's (collectively 8.95 million t per year; Fig. 6a), and has been lower and relatively stable since 1995 (collectively ~ 7.6 billion t per year). The peak in predation mortality of age 1 pollock in 2006 corresponds to the maturation of a large age class of 5-7 year old pollock and 2 year old Pacific cod that dominated the age composition of the two species in 2006. Increased temperatures were associated with ~6% increases in $M2_{i1,y}$ of pollock and ~15% and ~12% increases in $M2_{i1,y}$ of Pacific cod and arrowtooth flounder, respectively.

Pollock are both the dominant predator and a primary prey species in the multi-species model, second only to the 'other prey' category (Fig. 6a, b). After 'other prey' and pollock, the next most dominant prey category consumed is Pacific cod, followed by arrowtooth flounder (Fig. 6b). Pollock are primarily consumed by older conspecifics, and pollock cannibalism accounted for 51% (on average) of total predation mortality for age

1 pollock until 2006 when predation by arrowtooth flounder exceeded cannibalism as the largest source of predation mortality of age 1 pollock since 2007; Fig. 7).

The multi-species version of CEATTLE appears to compensate for elevated predation mortality on younger age classes by increasing estimates of recruitment. Thus, recruitment is higher in the multi-species model than in the single-species model for all three species, especially those with high predation rates (i.e., pollock). The direction of change in annual recruitment estimates from year-to-year was generally the same for both models (i.e., both models increased or decreased recruitment in the same year) with a notable exception; the multi-species model predicted a drop in recruitment between 2011 and 2012 for arrowtooth flounder, whereas the single-species model estimated very little change in recruitment for arrowtooth flounder between these years (Fig. 8c). Pollock recruitment from the single-species version of CEATTLE was positively correlated with Pacific cod recruitment ($R^2 = 0.75$) and inversely correlated with arrowtooth recruitment ($R^2 = -0.38$). However, correlations between pollock recruitment and Pacific cod or arrowtooth recruitment were weaker in the multi-species model ($R^2 = 0.60$ and -0.36, respectively).

The single- and multi-species models estimate similar fishing mortality rates for pollock that have remained relatively stable since the early 1980's (0.6-0.9% annual increase since 1982), with only a slight increase in recent years (Fig. 9). Both models also predict low and slightly decreasing fishing mortality rates for arrowtooth flounder (~2.1% annual decline since 1980). However, the two models yield somewhat divergent patterns of fishing mortality for Pacific cod over time, with single- and multi-species models

indicating a 10.8% and 14.5% increase (respectively) in fishing mortality since 1982 (Fig. 9).

3.2 Harvest scenarios and reference points

Projecting CEATTLE forward under mean recruitment produces trajectories of female spawning stock biomass that can be used to derive multi-species biological reference points and attendant fishing mortality rates. For species with high rates of cannibalism (e.g., pollock), projections under mean recruitment lead to oscillations in biomass similar to over-compensation recruitment dynamics (Fig. 10; *sensu* Botsford, 1986). However, a 50-year projection period was sufficient to allow such dynamics to reach a relative equilibrium under mean recruitment (Fig. 10 and S4).

In general, unfished and harvested female spawning stock biomass ($\overline{SSB}_{0,i}$ and $\overline{SSB}_{F,i}$, respectively) were lower for the multi- than the single-species model (with the exception of harvest scenario 4.3; Fig. 10 and S4). Unfished female spawning biomass from the multi-species version of CEATTLE was also lower than the model estimates of historical female spawning biomass for pollock, higher than historical female spawning biomass for Pacific cod, and approximately equal to recent female spawning biomass for arrowtooth flounder (Fig. 10). For pollock, which experienced the highest predation mortality rates, unfished female spawning biomass ($\overline{SSB}_{0,i}$) varied slightly, but was generally consistent among harvest scenarios, whereas $\overline{SSB}_{F,i}$ varied markedly among scenarios.

Fishing predator populations relieves interspecific predation pressure on prey species and results in spawning stock biomass trajectories that are higher than those predicted with $F_i = 0$ for all species simultaneously, as is evidenced from differences

between harvest scenario 1.1 (i.e., defining $\overline{SSB}_{0,i}$ based on $F_i = 0$ for all species simultaneously) and harvest scenarios 1.3 and 1.4 (i.e., $F_i = 0$ sequentially; Fig. 10). In some cases, harvested biomass of pollock was higher than unfished biomass because predator populations were reduced through harvest, and predation mortality was attenuated when predator populations were fished heavily to meet control rule objectives, (e.g., ~2025 values from harvest scenarios 3.2, 3.3, and 4.3; Fig. 10).

Estimates of $\overline{ABC}_{x,i}$ for pollock were slightly higher for single- than multi-species models for harvest scenarios with individual species yield targets (scenarios 1 and 2; Fig. 11). The opposite pattern was observed for pooled yield targets, where multi-species yield exceeded single species yield (scenarios 3 and 4; Fig 11). The exception was the multi-species model estimates of pollock $\overline{ABC}_{x,i}$ for all harvest scenarios when projected pollock female spawning biomass was constrained to be above 35% of unfished female spawning biomass (i.e., harvest scenarios 3.2, 3.4 and 4.3). The multi-species model estimated slightly lower $\overline{ABC}_{x,i}$ values than the single species model for most harvest scenarios for Pacific cod and arrowtooth flounder (Fig. 11).

Our model results, including emergent reference points, differ both slightly, in the case of harvested spawning biomass, to markedly (for unfished biomass) from BRPs from the actual 2012 single species assessments of each species in the Bering Sea. Average multi-species CEATTLE model estimates of $\overline{SSB}_{F,i}$ across harvest scenarios 1-3 were less than those reported in the 2012 assessments (i.e., 0.63, 0.50, and 0.56 times the 2013 $SSB_{40,i}$ projected values reported in the 2012 assessments for pollock, Pacific cod, and arrowtooth flounder respectively; Fig. 11). For the individual proxy ABC biomass harvest scenario (1), which is probably most similar to current control rules for these

species in the Bering Sea, Pacific cod single-species $\overline{SSB}_{40,i}$ (~344,000 t) was similar to that from the 2012 assessment of 358,000 t (Thompson and Lauth 2012), whereas the remaining single-species $\overline{SSB}_{40,i}$ and all of the multi-species $\overline{SSB}_{40,i}$ were less than the Pacific cod and arrowtooth assessment estimates of 2013 $SSB_{40,i}$ (with the exception of harvest scenarios 3.3 and 3.4 from the single-species model). Under harvest scenarios that included minimum biomass thresholds above $SSB_{35,i}$ (i.e., 2.1, 3.2 and 4.3), the multi-species model produced pollock $\overline{SSB}_{40,i}$ (2,278,000 t) estimates that were approximately equal to the 2013 $SSB_{40,i}$ estimated by Ianelli et al. (2012; i.e., 2,570,000 t).

Harvest scenario (4) which solved for the maximum yield of an individual species (4.1), or all three species in aggregate (4.2 and 4.3), resulted in maximum yields from the multi-species model for all three species combined of 2.7 and 3.2 million t (4.1 and 4.2, respectively), but only 0.46 million t if minimum biomass thresholds were included (i.e., individual populations were to remain above 35% of unfished female spawning biomass; 4.3; Fig. 11). Maximizing aggregate yield (4.2) resulted in depleting predator species (Pacific cod and arrowtooth) to extinction to minimize pollock predation and maximize pollock harvest, unless minimum biomass thresholds were implemented (i.e., 4.3, in which case predator species were depleted to the minimum limit).

Although temperature did not significantly alter estimates of recruitment and mortality in retrospective analyses, inclusion of temperature in projections did have a strong effect on BRP and MBRP estimates including $ABC_{x,i,y}$ (Fig. 12). Variability in $ABC_{x,i,y}$ introduced from temperature-driven changes to weight-at-age alone (i.e., single-species models, left panels of Fig.12) was as great as variability from various harvest

scenarios for single species models, while variation in $ABC_{x,i,y}$ due to harvest scenarios greatly exceeded temperature effects in multi-species models. Additionally, accounting for temperature effects in projections biased $ABC_{x,i,y}$ higher in single species models compared to the same models without temperature effects, a pattern not observed in multi-species models for pollock and Pacific cod.

4. Discussion

We found three patterns in reference points from projected harvest scenarios of CEATTLE that may be of broad interest. First, we observed differences in spawning biomasses at the ABC proxy between harvest scenarios and single- and multi-species models, yet discrepancies in spawning stock biomass estimates did not translate to large differences in yield (i.e., long term $\overline{ABC}_{x,i}$, $ABC_{x,i,y}$ averaged for the last five years of the simulation period). Multi-species estimates of $\overline{SSB}_{F,i}$ were 5-88% lower than single-species estimates of $\overline{SSB}_{F,i}$, and estimates of $\overline{SSB}_{F,i}$ varied around 9% among harvest scenarios for pollock in the multi-species CEATTLE model (17% and18% for Pacific cod and arrowtooth, respectively), yet $\overline{ABC}_{x,i}$ only varied slightly between harvest scenarios or models. This pattern suggests that $\overline{ABC}_{x,i}$ estimated from CEATTLE through this approach is robust to control rule specification and multi-species parameterization. The exception being pollock $\overline{ABC}_i$ from harvest scenarios 3.1, 3.3, 4.1, and 4.2 (i.e., those that were not constrained with a biomass threshold), where $\overline{ABC}_{x,i}$ from the multi-species model greatly exceeded that of the single-species model and the 2013 assessment estimates of ABC.

Secondly, we found that multi-species models produced higher estimates of

combined yield for aggregate MSY targets (harvest scenarios 3 and 4) than single species models, but were more conservative than single-species models when individual MSY targets were used (scenarios 1 and 2). The exception was scenario 4.1 and those with minimum biomass thresholds, 3.2, 3.4 and 4.3. Multiple previous studies have reported lower multi-species MSY as compared to aggregate single-species MSY; Guillen et al. (2013) found that MSY estimates from their multi-species models were 0.4% -2.7% lower than single-species model estimates and Gaichas et al. (2012) reported MSY from multi-species models was lower than the sum of individual single-species estimates of MSY. Tyrrell et al. (2011) suggested that multi-species models produce biological reference points that are more conservative than single-species model estimates, but our results challenge this universality and support earlier findings by Collie and Gislason (2001) that differential changes in BRPs with and without predation/cannibalism are dependent on aggregate versus species-specific targets.

Our results are also consistent with a recent study that reported higher estimates of MSY from multi-species than single-species Atlantis models, a pattern the authors attributed to increased biomass of prey (i.e., forage fish) partially released from predation by harvest of predator species (Smith et al., 2015). Gamble and Link (2012) noted a similar pattern in their multi-species model of the Northwestern Atlantic, where species affected by strong trophic interactions often exhibited competitive or predatory release when other species' harvest rates were increased. Annual predation on juvenile pollock in our model was 2-5 times the annual harvest of adult fish in the system, thus predators represent a strong control on population dynamics that is attenuated through harvest driven reductions in predator populations. Indeed in simulations without fishing,

increased predation mortality reduced unfished female pollock spawning biomass (relative to single-species estimates) in several harvest scenarios to levels sometimes below harvested spawning biomass. Such compensatory dynamics are not reflected in the single species model, nor fully exploited in individual species target scenarios (scenarios 1 and 2) that produced lower (arguably more conservative) estimates of ABC.

Lastly, our multi-species model results suggest harvest rates should be slightly lower than single-species corollaries when a management objective is to maintain a population above the MSY biomass proxy. This is particularly true when maximizing aggregate yield (e.g., as might be the case in a mixed species fleet); maximizing yield (harvest scenarios 4.1 and 4.2) resulted in fishing pollock predators (Pacific cod and arrowtooth flounder) to extinction (e.g., Fig. 10). Multiple extinctions under the maximum yield harvest scenario 4, was an undesired, but not necessarily unexpected outcome given the divergent productivities of the three species in the model. Gaichas et al. (2012) observed a similar pattern in their simulation of BRPs from surplus production models of multiple species complexes from temperate systems. They found that species complexes best able to balance yield and biodiversity objectives were those that combined species with similar productivity rates, environmental sensitivity and species interactions. In contrast, for species with divergent life history traits, aggregate BRPs often sacrificed biodiversity for yield, leading to multiple extinctions within a species complex. We found that constraining harvest scenario 4 to maintain the ratio of fished and unfished spawning stock biomass above the current MSY proxy ($0.35\ \overline{SSB}_{0,i}$) allowed the model to maximize aggregate yield while maintaining sustainable populations. In fact, all of the harvest scenarios that included a minimum biomass

threshold (i.e., 1.2, 3.2, 3.4, and 4.3) produced higher estimates of spawning stock biomass and lower estimates of yield than respective scenarios without biomass thresholds. Moreover, three harvest scenarios (3.2, 4.1, and 4.2) produced estimates of ABC that would exceed the current 2.0 million t optimum yield cap on groundfish harvest in the Bering Sea and Aleutian Islands, above which fishery harvest of all groundfish species immediately ceases (NPMFC, 2013). Thus we suggest that in addition to minimum biomass thresholds, implementation of CEATTLE for EBFM in Alaska could also include a maximum yield limit (i.e., combined 2 million t annual maximum) to meet EBFM goals of the North Pacific Fishery Management Council.

Similar to previous authors, we found that reference point values, in particular unfished biomass ($SSB_{0,i}$), are highly dependent on the background harvest rates of the other species in the model when reference points are determined (e.g. Collie and Gislason, 2001, Smith et al., 2015; Curti et al., 2013; Tyrell et al., 2011) and as such the number of possible values for a species' MBRP increases exponentially with increasing species interactions. However, differences in MBRPs will only be appreciable for those species that exhibit strong trophic interactions (direct and indirect) thus the potential number of MBRPs can be rapidly reduced *a priori* based on known trophic relationships. In our study, fishing was simultaneously set to 0 in harvest scenario 1.1 for all three species, $SSB_{0,i}$ of pollock prey was much lower than in harvest scenarios 1.3 and 1.4 when predator populations were assumed to continue to be harvested while pollock prey were unfished (e.g., as might be the case in separate or foreign fleets for each species), yet similar sensitivity to "fishing order" was not seen for Pacific cod and arrowtooth flounder. This model behavior adds a level of complexity to application of MBRPs

towards fisheries management, but does not represent an insurmountable challenge. In fact, using this projection approach for deriving predator (or prey) dependent MBRPs allows multi-species models to be readily utilized in trade-off analyses, especially with respect to multi-species fleets, conflicting conservation and extractive objectives, and harvest strategies for fisheries with high non-target species bycatch rates. Further extensions of the harvest scenario approach for deriving MBRPs could also include evaluation of differential effects of bottom-up drivers of productivity on predator and prey populations, and evaluation of jurisdictional borders and area closures that protect either (or both) prey and predator populations from fishing. Despite similar performance and model fits, the multi-species model estimated higher recruitment (all species) and lower female spawning biomass (pollock) than single-species models. This is consistent with findings of previous authors; Gislason (1999), Collie and Gislason (2001), Jurado-Molina et al. (2005), Kinzey and Punt (2009), and Van Kirk (2010) all reported higher recruitment estimates from their multi-species than single species models. Hollowed et al. (2000b) and Kinzey and Punt (2009) also observed lower spawning stock biomass for species in their multi-species versus single-species models, and Gislason (1999) estimated higher spawning stock biomass for herring (*Clupea harengus*) and sprat (*Sprattus sprattus*) in multi-species than single species VPA, while that of predatory cod (*G. morhua*) was similar between models. Further, trophic interactions constrained population abundances during multi-species projections (especially for pollock) under no-fishing scenarios even without stock recruitment relationships (i.e., under mean recruitment). We also observed decreasing oscillations in pollock biomass over time similar to over-compensation recruitment dynamics (i.e., “transitory disturbance”;

Beverton and Holt, 1957; Botsford 1986; e.g., Fig. 10 and S4). Both of these patterns appear due to cannibalism on ages 1-3 in the multi-species model, which reduces the magnitude of large cohorts and contributes to biomass variability. This is consistent with previous studies; Baily (1989) found that 30% of the diets of older pollock contained juvenile conspecifics, and that cannibalism was higher during daytime diurnal vertical migrations from surface waters to depths preferred by older conspecifics. Similarly, Boldt et al., 2012 reported increased rates of cannibalism in areas where small and large pollock overlap, as well as a decrease in pollock cannibalism between 1982-2006 due to reduced numbers of pollock prey and large cannibalistic conspecifics. We did not observe the latter pattern per se; we observed a slight drop in predation mortality from peak values between 1983-1993, and a sharp decline (starting around 2005) in the relative contribution of cannibalistic conspecifics to pollock predation mortality. The decline in pollock cannibalism corresponds to a marked increase in arrowtooth flounder predation that has exceeded that of other predators in the model to become the largest source of predation mortality since 2005. These results support recent postulations that increasing trends in arrowtooth biomass (combined with increasing prey demand under warming conditions) could negatively impact future pollock populations in the Bering Sea (e.g., Zador et al., 2011; Hunsicker et al., 2013).

Finally, we estimate predation mortality on juvenile pollock that is relatively high (around 1.8), but is consistent with values reported elsewhere for prey forage species or fish with high rates of interspecific predation and cannibalism. Curti et al. (2013) reported M2 values of 0.2-1.0 for silver hake (*Merluccius bilinearis*) and Atlantic herring (*Clupea harengus*) from the Georges Bank in their model, Tsechaye et al. (2014) report

predation mortalities for 1 year old Chinook salmon (*Oncorhynchus tshawytscha*) that ranged between of 0.3-1.62, Collie and Gislason (2001) used a mortality rate of 2.2 for their upper range of cannibalism in their simulations, Kinzey and Punt (2009) report M2 as high as 5.0 and 2.0 for 0+ and 1-year-old pollock in the Aleutian Islands, respectively. Lastly, Jurado-Molina et al. (2005), using a subset of our data and a two species MSCAA model for pollock and cod reported M2 values of 0.2-1.2 for 1+ pollock.

The CEATTLE model has some advantages over previous MSCAA modeling efforts in the Bering Sea, most notably (1) projection of the model under various control-rule scenarios (*sensu* Moffit et al., this issue) and (2) quasi-mechanistic inclusion of temperature-specific growth and predation rates. This provides an approach to evaluate the direct effects of climate-driven changes to thermal experience (i.e., increased metabolic demand and concomitant predation pressure) under future climate change on mortality, age composition, and reference points. Inclusion of temperature in CEATTLE contributed to differential estimates of recruitment between single- and multi-species models, especially for pollock where age 1 M2 increased during warm years in the mid 2000's. Single- and multi-species model projections also show that the indirect effect of temperature on weight-at-age influences future biomass estimates even before temperature effects on predator rations and concomitant predation rates are also included; weight-at-age affects all aspects of the model from fisheries yield to predator rations and is therefore a potentially sensitive component of CEATTLE. That said, the direct effects of temperature on the model were detectable, but much less influential than changes in predation (as evidenced from comparisons between models fit to observed temperatures versus mean historical temperatures). Collie and Gislason (2001) identified a similar

pattern where BRPs were much more sensitive to changes in natural mortality than changes in growth, and led them to suggest that fishing mortality rates should be conditioned on the level of predation mortality. Ianelli et al. (this issue) projected single and multi-species model under various climate scenarios and found that variation in recruitment estimates and harvest rates was larger than climate-driven changes to growth and predation. We similarly found that future MBRPs were more sensitive to changes in biomass targets and harvest scenarios than temperature-induced variability from projections.

Yet, even small changes in temperature can influence interspecific synchrony, compensation dynamics, and food web stability in ways that can be difficult to detect and predict (e.g., Gouhier et al., 2010), and accounting for temperature-driven changes to growth and predation is an important step towards disentangling direct and indirect climate effects on fish and fisheries. This is demonstrated by higher $ABC_{x,i,y}$ for single species models with temperature effects than the same model without temperature effects, which is not observed in the multi-species models for pollock and Pacific cod. Multiple authors have projected increases in fish biomass associated with increased temperature-driven growth rates (e.g., Cheung et al., 2015), but our results suggest that increases in trophic demand may counter direct effects of increasing temperatures on growth and production. Such interactions warrant further evaluation as they denote disparate changes in projected harvest under future climate change, but to do so will require additional modifications to CEATTLE that are beyond the scope of this study.

The temperature-specific VBGF in the model presently uses a covariate for temperature to annually modify the *d* parameter (i.e., the allometric slope of

consumption), but it is not fully mechanistic. Apparent changes in weight-at-age with temperature reflect both the direct physiological relationship of growth with temperature (typically dome-shaped, with a thermal optimum; Hanson et al., 1997; Holsman and Aydin, 2015), climate-driven shifts to prey quality and quantity (e.g., Heinz et al., 2013), and annual variation in size-selective mortality that may co-vary with temperature (e.g., Coyle et al., 2011; Hunt et al., 2011; Heinz et al., 2013). Our current approach could be improved by predicting growth in CEATTLE from bioenergetic algorithms used to predict consumption for an individual predator (e.g., Morita et al., 2010; Siddon et al., 2013). This should not alter results in the estimation phase, but could add a level of realism as well as "bottom-up" dynamics to the model under various future climate scenarios.

Additionally, numerous reviews of multi-species models suggest that they can improve estimates of natural mortality and recruitment and clarify mechanisms of spawner–recruit relationships (see Hollowed et al., 2000a; Link, 2010; Link and Browman, 2014). We similarly envision contrasting single- and multi-species spawner–recruit relationships to further retrospectively evaluate interacting bottom-up and top down trophodynamics believed to influence groundfish recruitment in the Bering Sea (sensu Mueter et al., 2007, 2011; Coyle et al., 2011). A logical extension of our present work could involve model projections using recruitment estimated from spawner-recruitment relationships with and without environmental covariates (see overviews in Link, 2010; Link and Browman, 2014; Fogarty, 2014; Plagányi et al., 2014). Combining such an approach with a more explicit bioenergetics-based VBGF as suggested above, would help evaluate indirect environmental effects on groundfish recruitment in the

Bering Sea and clarify interacting climate and fishery effects on recommended harvest rates.

Our intention was to evaluate the ability of CEATTLE to produce environmentally-conditioned reference points for fisheries management, but a few additional advancements could help refine our approach. We included diet information as known data, yet statistically fitting suitability coefficients might clarify predator-prey trophodynamic relationships in the model, and could be used to test hypotheses of trophic interactions under various climate and harvest conditions. Kempf et al. (2010) used a similar approach in their MSCAA model, and found that inclusion of predator and prey overlap in models affected recovery trajectories of North Sea fish populations under future simulations. Finally, we evaluated a range of harvest scenarios to demonstrate a diversity of reference points that could be estimated using this approach. The harvest scenarios included in this analysis are by no means exhaustive, and a number of other scenarios that include subtle or substantial variations of the harvest scenarios explored here could be evaluated. Future applications that include evaluation of $B_{trigger}$ sloping control rules, maximum yield limits, and bycatch constraints would further advance the use of CEATTLE for use in a tactical EBFM context.

5. Application of MBRPs toward EBFM

Development of diverse MPRBs from multi-species models is a necessary step in moving forward with EBFM (Link, 2010; Link and Browman, 2014). Projecting CEATTLE provides proxies for MBRPs that can readily be implemented in current OFL control rules for Alaska fisheries management and demonstrates the range of possible

considerations as well as individual strengths and weaknesses of each control rule approach. Like previous authors, we found that ABC proxies were lower than the single-species CEATTLE model estimates for all harvest scenarios where minimum biomass thresholds were included in control rules (e.g., Gaichas et al., 2012). Although implementation of control rules will depend on regional management objectives, our results can provide some generalized EBFM guidance going forward. First, MBRPs are not inherently more conservative than single-species corollaries (i.e., BRPs), comparative risk depends on the degree of inter-specific predation and cannibalism, yet we found that recommended harvest rates were relatively consistent between harvest scenarios, especially if target biomasses are included for individual species. Second, climate and trophic drivers can interact to affect MBRPs, but for prey species with high predation rates, trophic and management-driven changes may exceed direct effects of temperature on growth and predation. Given this, MSCAA models can readily be used for tactical EBFM decisions under changing climate conditions, if, as suggested here and by various authors previously, harvest scenarios used for deriving MBRPs combined a minimum biomass threshold with yield targets to meet biodiversity and yield objectives (Worm et al., 2009; Gaichas et al., 2012).

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Figure 1. Conceptual figure of the Climate-Enhanced, Age-based model with Temperature-based Tropic Linkages and Energetics (CEATTLE model) for prey species *i*, age *j* in year *y*. See table 3 for parameter definitions.

Figure 2. Eastern Bering Sea bottom temperature (^{o}C) values used for CEATTLE model. Mean bottom temperatures from survey observations (solid line) and from GAM regressions of bottom temperature (dashed lines) based on estimates of SST from regional downscaled IPCC projections (see Ianelli et al., 2011 for methods). The vertical dashed line represents the start of the projection period. Shading represents +/- 1 standard deviation of the mean.

Figure 3. Single- (gray lines) and multi-species (black lines) retrospective model estimates of total (thick solid lines), female spawning (dashed lines), and survey biomass (thin solid lines). Filled circles represent mean observed groundfish survey biomass and standard errors of the mean (error bars).

Figure 4. Temperature effects on model estimates of survey biomass and percent difference in model estimates with temperature versus models without. Single species models with temperature (SS with T); multispecies models with Temperature (MS with T), single species models where temperature was held constant at 2.51 (SS no T), multispecies models with temperature was held constant at 2.51 (MS, no T).

Figure 5. Annual variation in total mortality ($M1_{i1}$+$M2_{i1,y}$) for age 1 pollock (a), Pacific cod (b), and arrowtooth flounder (c) from the single-species models (dashed line), multi-species models with temperature (black line), and multi-species models where temperature was held at a constant mean observed value of 2.51 ^{o}C (gray line).

Figure 6. a) Combined total predator ration (all three predators combined) over time grouped by predator. b) Total prey consumed by all three predators combined (note the log scale). c) Pollock M2 (age 1 only) consumed by each predator species.

Figure 7. Proportion of total predation mortality for age 1 pollock from pollock (solid), Pacific cod (dashed), and arrowtooth flounder (dotted) predators across years.

Figure 8. Annual single- and multi-species CEATTLE model estimates of recruitment (age 1) for pollock (a), Pacific cod (b), and arrowtooth flounder (c). Lighter shading represents the 95% CI around mean estimates. Darker shading represents +/- 1 standard error of the mean estimate.

Figure 9. Time-trajectories of single- and multi-species (gray and black, respectively) CEATTLE model estimates of fishing mortality rate for eastern Being Sea walleye pollock, Pacific cod, and arrowtooth flounder (note that the single- and multi-species lines for arrowtooth flounder overlap).

Figure 10. Multi-species CEATTLE model projections of unfished (gray lines) and fished (black lines) spawning stock biomass (SSB_0 and SSB_{40}, respectively) for pollock under each harvest control rule (see Methods and Table 5 for control rule definitions).

Figure 11. Single- and multi-species CEATTLE model estimates of mean female spawning biomass at 40% of unfished biomass (i.e., a proxy for spawning biomass corresponding to Acceptable Biological Catch; SSB_{ABC}) and yield (proxy for ABC for control rules 1-3; proxy for MSY for control rule 4) for the last 5 years of the projection period for each species and harvest control rule. Error bars represent standard error of the mean across the five year period. For the left panels, dashed and solid horizontal lines represent the 2013 B_{40} and B_{35}, respectively, from 2012 stock assessments for each species (Ianelli et al., 2012, Thompson and Lauth 2012, Spies et al., 2012). For the right panel, dashed and solid horizontal lines represents the 2012 stock assessment maxABCs and OFLs, respectively, for each species.

Figure 12. Single- and multi-species projections of annual (y) yield ($ABC_{x,i,y}$) under harvest scenarios with individual species i targets (i.e., x scenarios 1-2). Shaded gray areas contain 95% of $ABC_{x,i,y}$ values from 5 different temperature projections. The solid line indicates the value from models fit and projected under a constant historical temperature (i.e., 5.21 ^{o}C).

Table 1. Population dynamics equations for species i and age j in each simulation year y. BT indicates the AFSC bottom trawl survey and EIT represents the echo-integrated acoustic-trawl survey. For all other parameter definitions see Table 3.

Definition	Equation		
Recruitment	$N_{i1,y} = R_{i,y} = R_{0,i}e^{\tau_{i,y}}$	$\tau_{i,y}\sim\mathrm{N}(0,\sigma^2)$	T1.1
Initial abundance	$N_{ij,1} = \begin{cases} R_{0,i}e^{(-j\,M1_{ij})}N_{0,ij} & y=1 \quad 1<j\le A_i \\ R_{0,i}e^{(-j\,M1_{i,A_i})}N_{0,i,A_i}/\left(1-e^{(-j\,M1_{i,A_i})}\right) & y=1 \quad j>A_i \end{cases}$		T1.2
Numbers at age	$N_{i,j+1,y+1} = N_{ij,y}e^{-Z_{ij,y}}$ $N_{i,A_i,y+1} = N_{i,A_i-1,y}e^{-Z_{i,A_i-1,y}} + N_{i,A_i,y}e^{-Z_{i,A_i,y}}$	$1\le y\le n_y \quad 1\le j<A_i$ $1\le y\le n_y \quad j>A_i$	T1.3
Catch	$C_{ij,y} = \frac{F_{ij,y}}{Z_{ij,y}}\left(1-e^{-Z_{ij,y}}\right)N_{ij,y}$		T1.4
Total yield (kg)	$Y_{i,y} = \sum_j^{A_i}\left(\frac{F_{ij,y}}{Z_{ij,y}}\left(1-e^{-Z_{ij,y}}\right)N_{ij,y}W_{ij,y}\right)$		T1.5
Biomass at age (kg)	$B_{ij,y} = N_{ij,y}W_{ij,y}$		T1.6
Spawning biomass at age (kg)	$SSB_{ij,y} = B_{ij,y}\rho_{ij}$		T1.7
Total mortality at age	$Z_{ij,y} = \mathrm{M1}_{ij} + \mathrm{M2}_{ij,y} + F_{ij,y}$		T1.8
Fishing mortality at age	$F_{ij,y} = F_{0,i}e^{\varepsilon_{i,y}}s^{\mathrm{f}}_{ij}$	$\varepsilon_{i,y}\sim\mathrm{N}(0,\sigma^2_{\mathrm{F},i})$	T1.9
Weight at age (kg)	$W_{ij,y} = W_{\infty,iy}\left(1-e^{\left(-K_i(1-d_{i,y})(j-t_{0,i})\right)}\right)^{\frac{1}{1-d_{i,y}}}$		T1.10a
	$d_{i,y} = e^{(\alpha_{d,i,y}+\alpha 0_{d,i}+\beta_{d,i}T_y)}$		T1.10b
	$W_{\infty,iy} = \left(\frac{H_i}{K_i}\right)^{1/(1-d_{i,y})}$		T1.10c
BT survey biomass (kg)	$\hat{\beta}^s_{i,y} = \sum_j^{A_i}\left(N_{ij,y}e^{-0.5\,Z_{ij,y}}W_{ij,y}s^s_{ij}\right)$		T1.11
EIT survey biomass (kg)	$\hat{\beta}^{eit}_y = \sum_j^{A_1}\left(N_{1j,y}e^{0.5(-Z_{1j,y})}W_{1j,y}s^{eit}_{1j}q^{eit}_{1j}\right)$	(pollock only)	T1.12
Fishery age composition	$\hat{O}^f_{ij,y} = \frac{C_{ij,y}}{\sum_j C_{ij,y}}$		T1.13
BT survey age composition	$\hat{O}^s_{ij,y} = \frac{N_{ij,y}e^{0.5(-Z_{ij,y})}s^s_{ij}}{\sum_j\left(N_{ij,y}e^{0.5(-Z_{ij,y})}s^s_{ij}\right)}$		T1.14
EIT survey age composition	$\hat{O}^{eit}_{1j,y} = \frac{N_{1j,y}e^{0.5(-Z_{1j,y})}s^{eit}_{1j}q^{eit}_1}{\sum_j\left(N_{1j,y}e^{0.5(-Z_{1j,y})}s^{eit}_{1j}q^{eit}_1\right)}$	(pollock only)	T1.15

BT selectivity	$s_{ij}^{S} = \frac{1}{1+e^{\left(-b_i^S \cdot j - a_i^S\right)}}$		T1.16
Fishery selectivity	$s_{ij}^{\mathrm{f}} = \begin{cases} e^{\eta_{ij}} & j \le A_{\eta,i} \\ e^{\eta_{i,A_{\eta,i}}} & j > A_{\eta,i} \end{cases}$	$\eta_{ij} \sim \mathrm{N}(0, \sigma_{\mathrm{f},i}^2)$	T1.17
Proportion females	$\omega_{ij} = \frac{e^{-j\,M_{\mathrm{fem}}}}{e^{-j\,M_{\mathrm{fem}}} + e^{-j\,M_{\mathrm{male}}}}$		T1.18
Proportion of mature females	$\rho_{ij} = \omega_{ij}\phi_{ij}$		T1.19
Weight at age (kg)	$W_{\mathrm{i}j,y} = W_{ij,y}^{\mathrm{fem}}\omega_{ij} + \left(1-\omega_{ij}\right)W_{ij,y}^{\mathrm{male}}$		T1.20
Residual natural mortality	$M1_{ij} = M_{ij}^{\mathrm{fem}}\omega_{ij} + \left(1-\omega_{ij}\right)M_{ij}^{\mathrm{male}}$		T1.21

Table 2. Predation mortality (M2) equations for predators *p* of age *a*, and prey *i* of age *j*.

Definition	Equation	
Predation mortality	$M2_{ij,y} = \sum_{pa}\left(\frac{N_{pa,y}\delta_{pa,y}\bar{S}_{paij}}{\sum_{ij}(\bar{S}_{paij}B_{ij,y}) + B_p^{other}(1-\sum_{ij}(\bar{S}_{paij}))}\right)$	T2.1
Predator-prey suitability	$\bar{S}_{paij} = \frac{1}{n_y}\sum_{y}\left(\frac{\frac{\bar{U}_{paij}}{B_{ij,y}}}{\sum_{ij}\left(\frac{\bar{U}_{paij}}{B_{ij,y}}\right) + \frac{1+\sum_{ij}\bar{U}_{paij}}{B_p^{other}}}\right)$	T2.2
Mean gravimetric diet proportion	$\bar{U}_{paij} = \frac{\sum_y U_{paij,y}}{n_y}$	T2.3
Individual specific ration (kg kg^{-1} yr^{-1})	$\delta_{pa,y} = \hat{\varphi}_p \alpha_\delta W_{pa,y}{}^{(1+\beta_\delta)} f(T_y)_p$	T2.4
Temperature scaling algorithm	$f(T_y)_p = V^X e^{(X(1-V))}$	T2.5
	$V = (T_p^{cm} - T_y)/(T_p^{cm} - T_p^{co})$	T2.5a
	$X = (Z^2(1+(1+40/Y)^{0.5})^2)/400$	T2.5b
	$Z = ln(Q_p^c)(T_p^{cm} - T_p^{co})$	T2.5c
	$Y = ln(Q_p^c)(T_p^{cm} - T_p^{co} + 2)$	T2.5d

Table 3. Parameter definition (n is the number of parameters for estimated parameters only, value (Plk: Pollock; Cod: Pacific cod; Atf: Arrowtooth flounder both sexes; Atf_M: Arrowtooth flounder males; Atf_F: Arrowtooth flounder females), and source. I: Input parameter (assigned); M: model index; E: Estimated parameter; F: fixed parameter P: Derived quantity; D: Data.

Parameter	Definition	Type	Value	Source
y	Year	M	$[1,2,3 \ldots n_y]$	e
p	Predator	M	$[1,2,3 \ldots n_p]$	e
a	Predator age (years)	M	$[1,2,3 \ldots A_p]$	e
i	Prey	M	$[1,2,3 \ldots n_i]$	e
j	Prey age (years)	M	$[1,2,3 \ldots A_i]$	e
n_i	Number of prey species	I	3	e
n_p	Number of predator species	I	3	e
A_i	Number of prey ages	I	Plk: 12 Cod: 12 Atf : 21	e
A_p	Number of predator ages	I	Plk: 12 Cod: 12 Atf: 21	e
n_y	Number of simulation years	I	34	e
y_0	Start year	I	1979	e
$\hat{\varphi}_p$	Annual relative foraging rate (d yr^{-1})	I	Plk: 0.119 Cod: 0.041 Atf: 0.125	a
α_δ	Intercept of the allometric maximum consumption function (g g^{-1} yr^{-1})	I	Plk: 0.119 Cod: 0.041 Atf: 0.125	a
β_δ	Allometric slope of maximum consumption	I	Plk: -0.460 Cod: -0.122 Atf: -0.245	a
T_p^{cm}	Consumption maximum physiological temperature (^{o}C)	I	Plk: 15.00 Cod: 21.00 Atf: 34.13	a
T_p^{co}	Consumption optimum physiological temperature (^{o} C)	I	Plk: 10.00 Cod: 13.70 Atf: 19.60	a
Q_p^c	Max consumption parameter	I	Plk: 2.60 Cod: 2.41 Atf: 2.18	a
$R_{0,i}$	Mean Recruitment; n=[1,1,1]	E	≥ 0	e
$\tau_{i,y}$	Annual recruitment deviation; n=[34,34,34]	E	number	e
$N_{0,ij}$	Initial abundance; n=[11,11,20]	E	≥ 0	e
$F_{0,i}$	Mean fishing mortality; n=[1,1,1]	E	≥ 0	e
$\varepsilon_{i,y}$	Annual fishing mort. deviation; n=[34,11,20]	E	number	e
η_{ij}	Fishery age selectivity coef. ; n=[8,8,8]	E	number	e
b_i^s	Survey age selectivity slope; n=[1,1,1]	E	number	e
a_i^s	Survey age selectivity limit ; n=[1,1,1]	E	number	e
$d_{i,y}$	VBGF allometric slope of consumption	P	≥ 0	e

$W_{\infty,iy}$	VBGF max asymptotic weight (kg)	P	> 0	e
$_{ij}$	Proportion of mature females at age	P	$\in [0,1]$	e
$M1_{ij}$	Residual natural mortality	F	≥ 0	e, h
$\alpha 0_{d,i}$	Intercept for VBGF *d* parameter	F	Plk: -0.817 Cod: -0.375 Atf_M: -0.213 Atf_F: -0.340	d
$\alpha_{d,i,y}$	Annual intercept for VBGF *d* parameter	F	See S10	d
$\beta_{d,i}$	Temperature covariate for VBGF *d* parameter	F	Plk: 0.0090 Cod: 0.0045 Atf_M: -0.0057 Atf_F: -0.0115	d
K_i	VBGF energy loss constant (kg kg^{-1} yr^{-1})	F	Plk: 0.22 Cod: 0.45 Atf_M: 1.08 Atf_F: 0.38	d
H_i	VBGF assimilation constant (kg kg^{-d} yr^{-1})	F	Plk: 16.34 Cod: 9.30 Atf_M: 5.19 Atf_F: 5.90	d
$t_{0,i}$	VBGF age when $W_{ij,y}$= 0 (years)	F	Plk: 0.53 Cod: -0.16 Atf_M: -1.00 Atf_F: -0.28	d
S_{1j}^{eit}	EIT survey selectivity	F	$\in [0,1]$	c
M_i^{fem}	Female natural mortality		Plk*: NA Cod: 0.37 Atf: 0.35	c
M_i^{male}	Male natural mortality	F	Plk*: NA Cod: 0.37 Atf: 0.20	c
ω_{ij}	Female proportion of population	F	$\in [0,1]$	c
ϕ_{ij}	Age-specific maturity proportions	F	$\in [0,1]$	c
$C_{i,y}^{*}$	Observed total yield (kg)	D	≥ 0	f
$O_{ij,y}^{f}$	Observed fishery age comp.	D	$\in [0,1]$	f
$O_{ij,y}^{s}$	Observed BT age comp.	D	$\in [0,1]$	b
$O_{1j,y}^{eit}$	Observed EIT age comp.	D	$\in [0,1]$	g
$_{i,y}^{s}$	Observed BT survey biomass (kg)	D	number	b
$_{y}^{eit}$	Observed EIT survey biomass (kg)	D	number	g
T_y	Bottom temperature (^{o} C)	D	number	b
$U_{paij,y}$	Gravimetric proportion of prey in predator stomach	D	$\in [0,1]$	b
B_p^{other}	Biomass of other prey (kg)	D	≥ 0	h

* pollock age-specific M1 residual mortalities from the assessment were used (same values for male and females).

a. Holsman and Aydin 2015
b. Alaska Fisheries Science Center eastern Bering Sea bottom trawl survey
c. Stock assessments (Ianelli et al., 2012; Thompson and Lauth, 2012; Spies et al., 2012)

d. Supplemental materials, this study
e. This study
f. Fishery observer data
g. Alaska Fisheries Science Center echo-integrated acoustic trawl survey
h. Juarado Molina et al., 2005

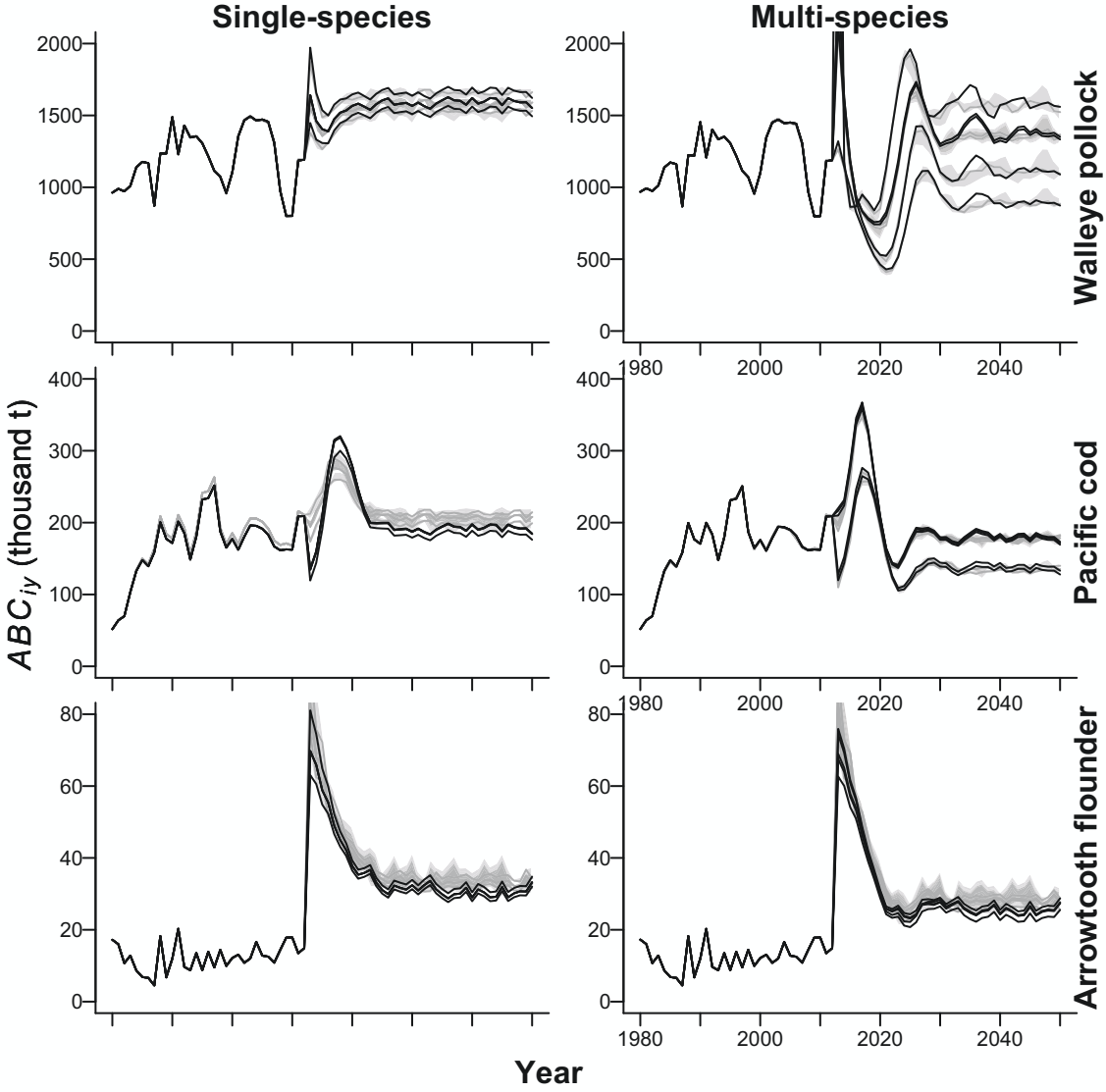
Table 4. Components of the likelihood function for each species *i* of age *j* in year *y*. EBS: Eastern Bering Sea. See Tables 2, 3 and 4 for parameter definitions.

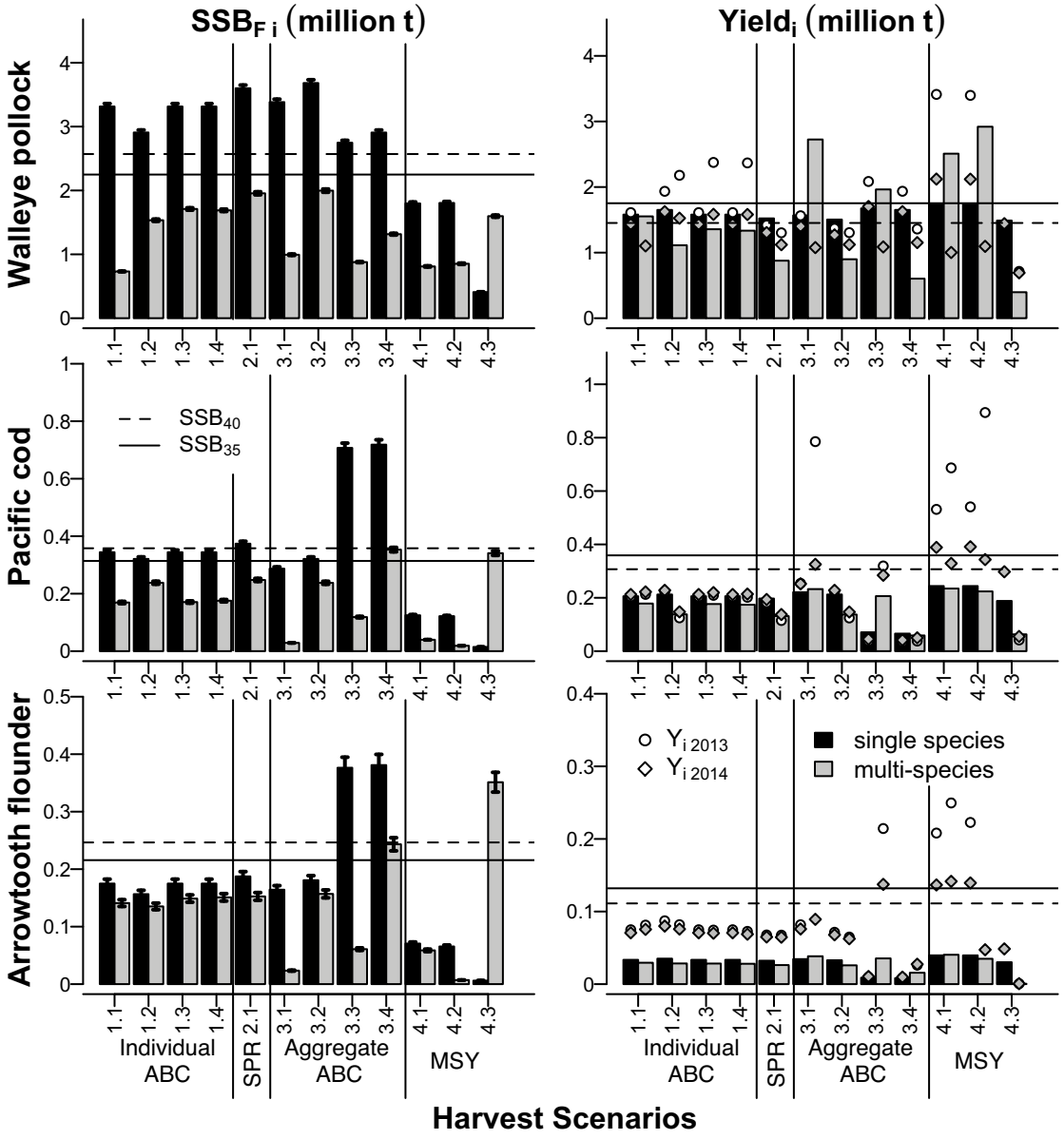
Description	Equation	Data source	
Data components			
BT survey biomass	$\sum_i \sum_y \frac{\left[ln\left(\beta^{s}_{i,y}\right)-ln\left(\widehat{\beta}^{s}_{i,y}\right)\right]^2}{2\sigma^2_{s,i}}$	NFMS annual EBS BT survey (1979-2012)	T4.1
BT survey age composition	$-\sum_i n_i \sum_y \sum_j \left(O^s_{ij,y}+v\right) ln\left(\hat{O}^s_{ij,y}+v\right)$	NFMS annual EBS BT survey (1979-2012)	T4.2
EIT survey biomass	$\sum_y \frac{\left[ln\left(\beta^{eit}_{y}\right)-ln\left(\widehat{\beta}^{eit}_{y}\right)\right]^2}{2\sigma^2_{eit}}, \sigma_{eit}=0.2$	Pollock acoustic trawl survey (1979-2012)	T4.3
EIT age composition	$-n\sum_y \sum_j \left(O^{eit}_{1j,y}+v\right) ln\left(\hat{O}^{eit}_{1j,y}+v\right)$	Pollock acoustic trawl survey (1979-2012)	T.4.4
Total catch	$\sum_i \sum_y \frac{\left[ln\left(C^*_{i,y}\right)-ln\left(\hat{C}^*_{i,y}\right)\right]^2}{2\sigma^2_C}, \sigma_C=0.05$	Fishery observer data (1979-2012)	T.4.5
Fishery age composition	$-\sum_i n_i \sum_y \sum_j \left(O^f_{ij,y}+v\right) ln\left(\hat{O}^f_{ij,y}+v\right)$	Fishery observer data (1979-2012)	T4.6
Penalties			
Fishery selectivity	$\sum_i \sum_j^{A_i-1} \chi \cdot \left[ln\left(\frac{\eta^{\mathrm{f}}_{ij}}{\eta^{\mathrm{f}}_{ij+1}}\right)-ln\left(\frac{\eta^{\mathrm{f}}_{ij+1}}{\eta^{\mathrm{f}}_{ij+2}}\right)\right]^2, \chi=\begin{cases}20, \text{if} & \eta^{\mathrm{f}}_{ij}>\eta^{\mathrm{f}}_{ji+1}\\ 0, \text{if} & \eta^{\mathrm{f}}_{ij}\le\eta^{\mathrm{f}}_{ij+1}\end{cases}$		T4.7
Priors			
	$\sum_i \sum_y \left(\tau_{i,y}\right)^2$		T4.8
	$\sum_i \sum_y \left(N_{0,ij}\right)^2$		T4.9
	$\sum_i \sum_y \left(\varepsilon_{i,y}\right)^2$		T4.10

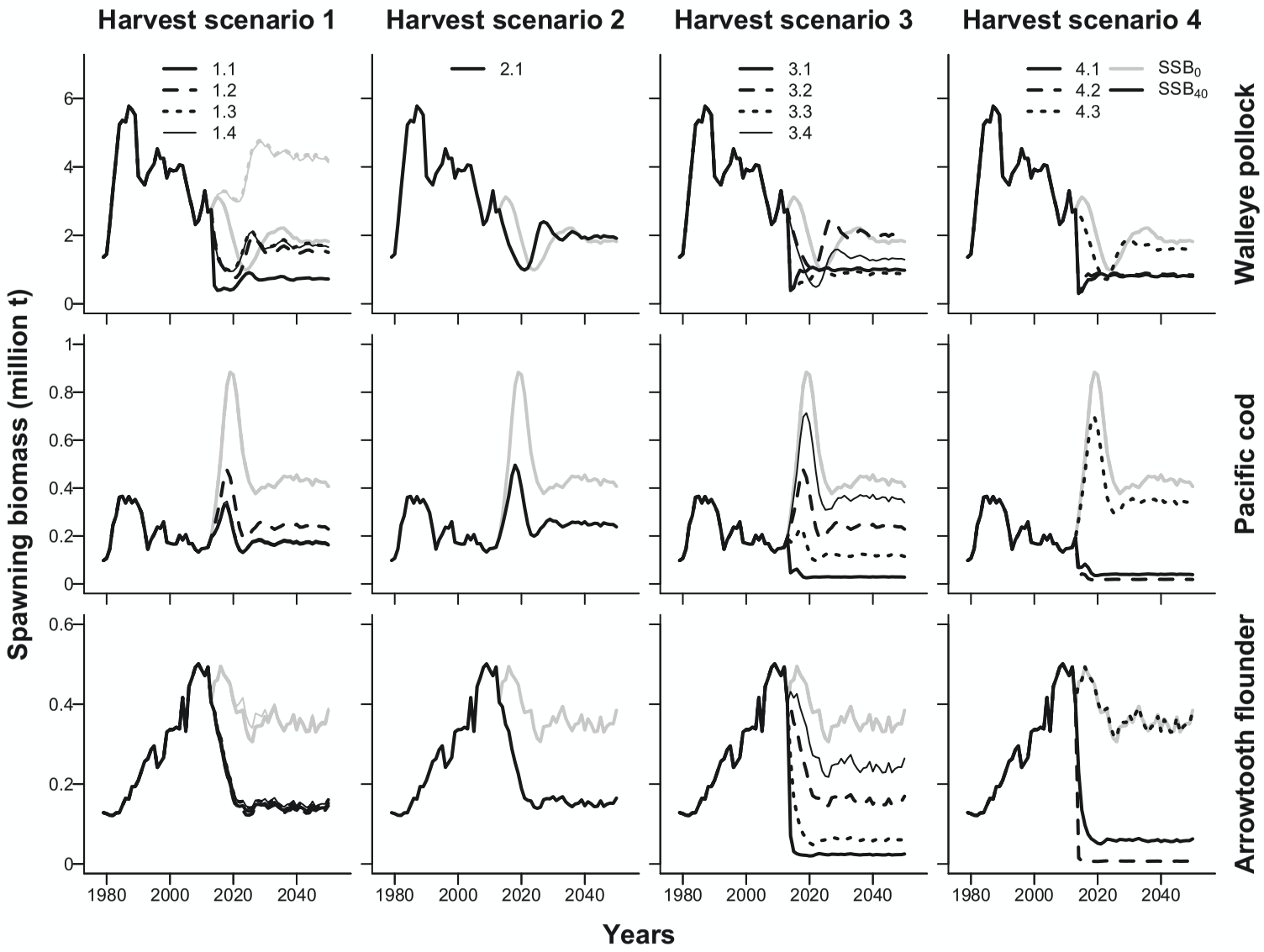
v=0.001

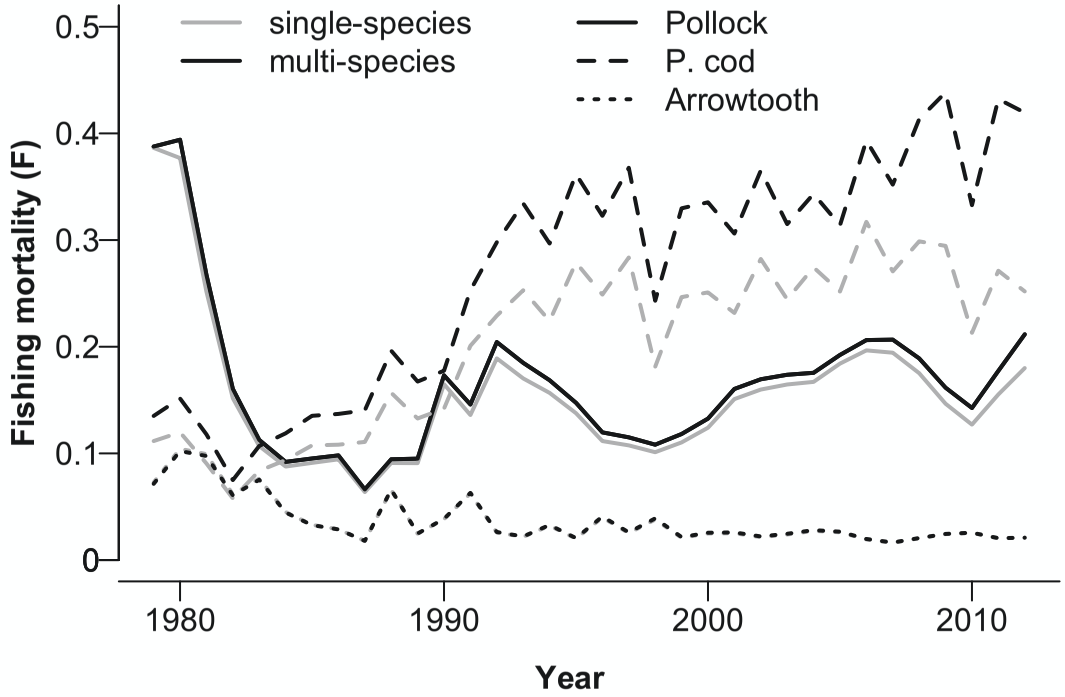
Table 5. Equations defining the application of the harvest control rules (CR).

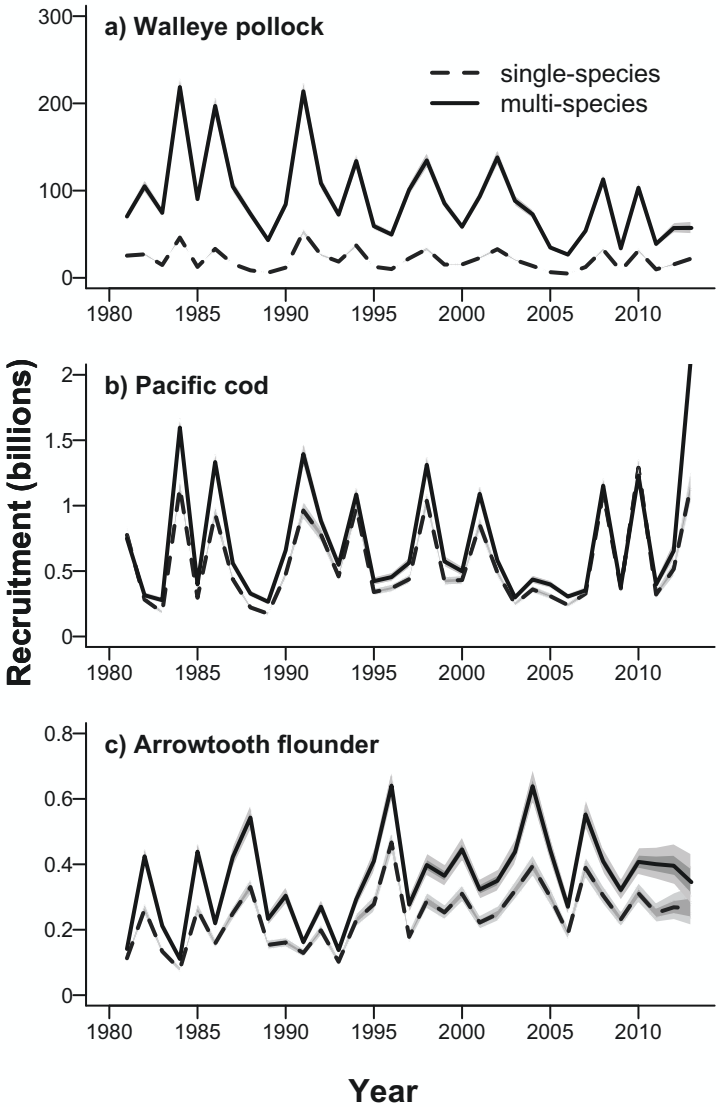
CR	Equation	
Biomass proxy for individual ABC		
1.1	$\overline{SSB}_{F,i} = 0.4\overline{SSB}_{0,i}$	T5.1.1
1.2	$\overline{SSB}_{F,i} = 0.4\overline{SSB}_{0,i} \cap SSB_{F,i} \geq 0.35 \cdot SSB_{0,i} \forall y \in n_y^{\text{fut}}$	T5.1.2
1.3	$\overline{SSB}_{F,i} = 0.4\overline{SSB}_{0,i}$ for $i = [2,3]$	T5.1.3a
	$\overline{SSB}_{F,1} = 0.4\overline{SSB}'_{0,1}$; $\overline{SSB}'_{0,1} = \left(\overline{SSB}_{0,1} \middle\| \overline{SSB}_{F,i} \forall i \in [2,3]\right)$	T5.1.3b
1.4	$\overline{SSB}_{F,2} = 0.4\overline{SSB}_{0,2}$	T5.1.4a
	$\overline{SSB}_{F,3} = 0.4\overline{SSB}'_{0,3}$; $\overline{SSB}'_{0,3} = \left(\overline{SSB}_{0,3} \middle\| \overline{SSB}_{F,2}\right)$	T5.1.4b
	$\overline{SSB}_{F,1} = 0.4\overline{SSB}'_{0,1}$; $\overline{SSB}'_{0,1} = \left(\overline{SSB}_{0,1} \middle\| \overline{SSB}_{F,i} \forall i \in [2,3]\right)$	T5.1.4c
Spawning biomass-per-recruit (SPR) approach		
2.1	$SPR_{F,i} = 0.4\left(SPR_{F,i} \middle\| F_{x,i} = 0\right)$	T5.2.1a
	$SPR_{F,i} = \frac{1}{R_{i,1}} \sum_j^{A_i} \left(\rho_{ij} \overline{W}_{ij} N_{ij} e^{-\left(F_{x,i} s^{\text{f}}_{ij} + M1_{ij} + \overline{M2}_{ij}\right)}\right)$	T5.2.1b
Biomass proxy for aggregate ABC		
3.1	$\sum_i \overline{SSB}_{F,i} = 0.4 \sum_i \overline{SSB}_{0,i}$; $F_{x,i} = F_{ALL}(\overline{M}_i / \sum_i \overline{M}_i)$	T5.3.1a
	$\overline{M}_i = \sum_y^{n_y} \sum_i^{A_i} \frac{M1_{ij} + M2_{ij,y}}{A_i n_y}$	T5.3.1b
3.2	$\sum_i \overline{SSB}_{F,i} = 0.4 \sum_i \overline{SSB}_{0,i} \cap SSB_{F,i} \geq 0.35 \cdot SSB_{0,i} \forall y \in n_y^{\text{fut}}$	T5.3.2a
	$F_{x,i} = F_{ALL_i} \frac{\overline{M}_i}{\sum_i \overline{M}_i}$	T5.3.2b
3.3	$\sum_i \overline{SSB}_{F,i} = 0.4 \sum_i \overline{SSB}_{0,i}$; $F_{x,i} = F_{ALL}\left(\overline{SSB}_{0,i} / \sum_i \overline{SSB}_{0,i}\right)$	T5.3.3
3.4	$\sum_i \overline{SSB}_{F,i} = 0.4 \sum_i \overline{SSB}_{0,i} \cap SSB_{F,i} \geq 0.35 \cdot SSB_{0,i} \forall y \in n_y^{\text{fut}}$	T5.3.4a
	$F_{x,i} = F_{ALL} \frac{\overline{SSB}_{0,i}}{\sum_i \overline{SSB}_{0,i}}$	T5.3.4b
Maximum yield		
4.1	$\max\left(\overline{ABC}_{x,i} \middle\| F_{x,i}\right) \forall i \in [1,2,3]$	T5.4.1
4.2	$\max\left(\sum_i \overline{ABC}_{x,i} \middle\| F_{x,i}\right)$	T5.4.2
4.3	$\max\left(\sum_i \overline{ABC}_{x,i} \middle\| F_{x,i}\right) \cap SSB_{F,i} \geq 0.35 \cdot SSB_{0,i} \forall y \in n_y^{\text{fut}}$	T5.4.3

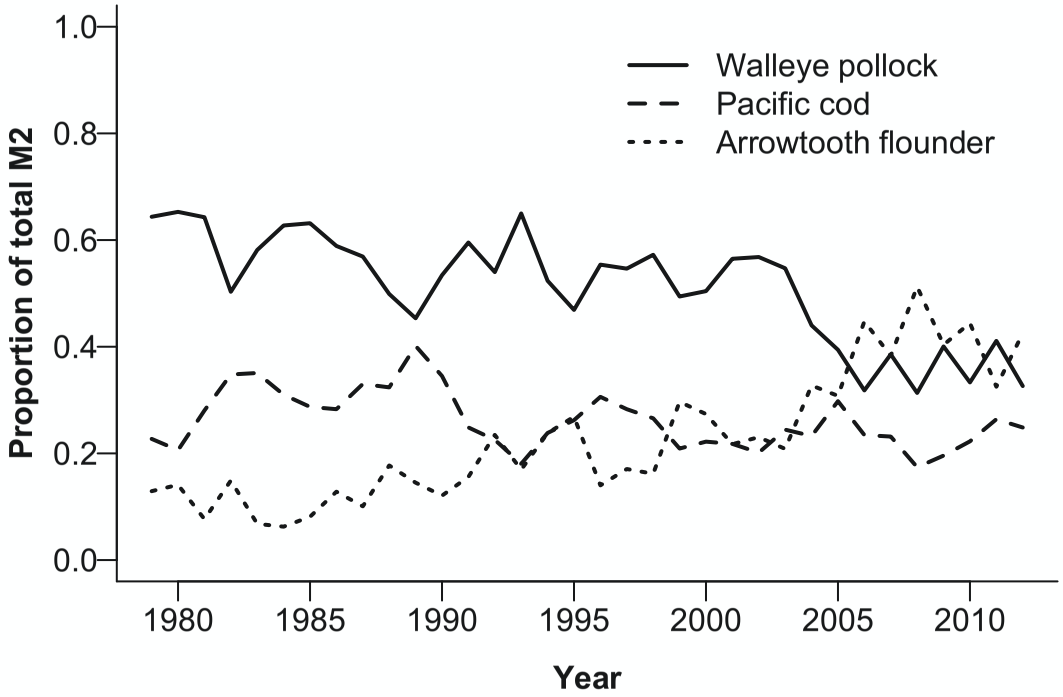

Single-species
Multi-species
Walleye pollock
Pacific cod
Arrowtooth flounder
ABC_{iy} (thousand t)
Year
2000
1500
1000
500
0
400
300
200
100
80
60
40
20
1980
2000
2020
2040

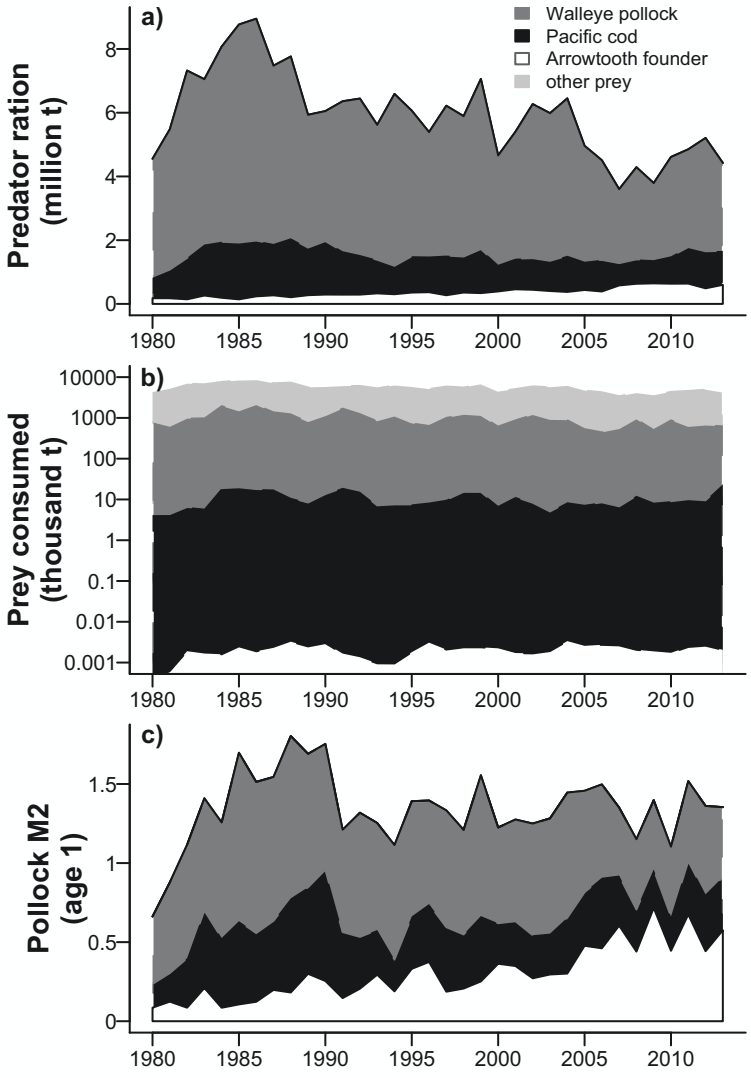

SSB$_{F\,i}$ (million t)
Yield$_i$ (million t)
Walleye pollock
Pacific cod
Arrowtooth flounder
SSB$_{40}$
SSB$_{35}$
Y$_{i\,2013}$
Y$_{i\,2014}$
single species
multi-species
1.1
1.2
1.3
1.4
2.1
3.1
3.2
3.3
3.4
4.1
4.2
4.3
Individual ABC
SPR
Aggregate ABC
MSY
Harvest Scenarios

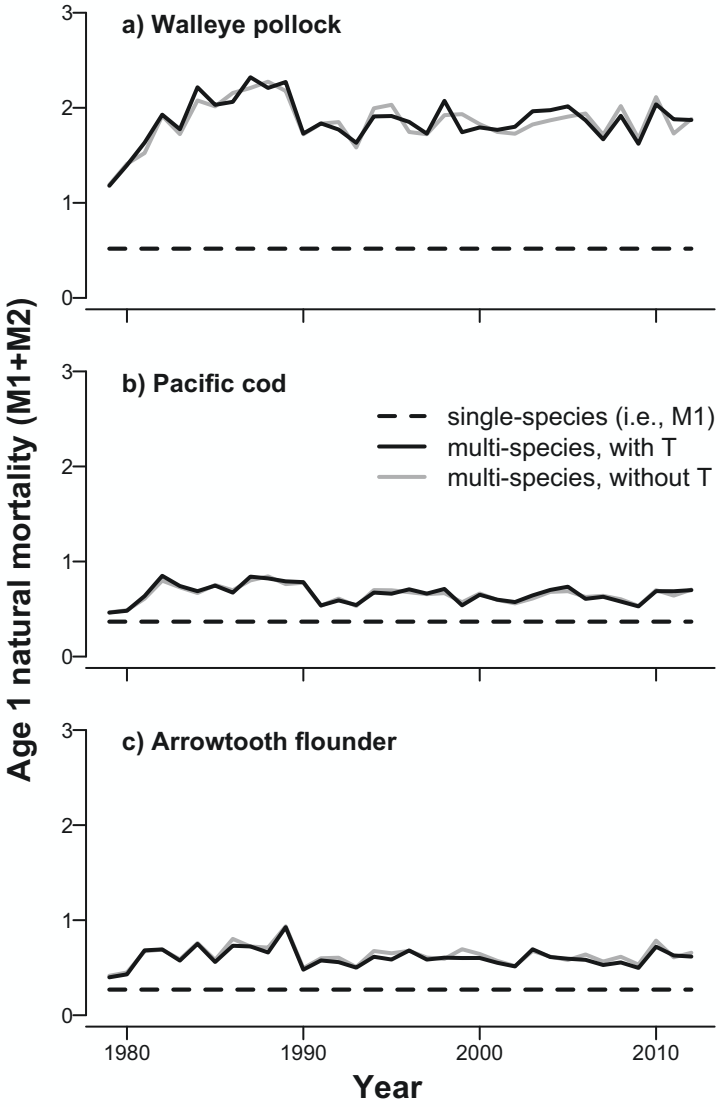

Harvest scenario 1
Harvest scenario 2
Harvest scenario 3
Harvest scenario 4
1.1
1.2
1.3
1.4
2.1
3.1
3.2
3.3
3.4
4.1
4.2
4.3
SSB0
SSB40
Walleye pollock
Pacific cod
Arrowtooth flounder
Spawning biomass (million t)
Years
1980
2000
2020
2040

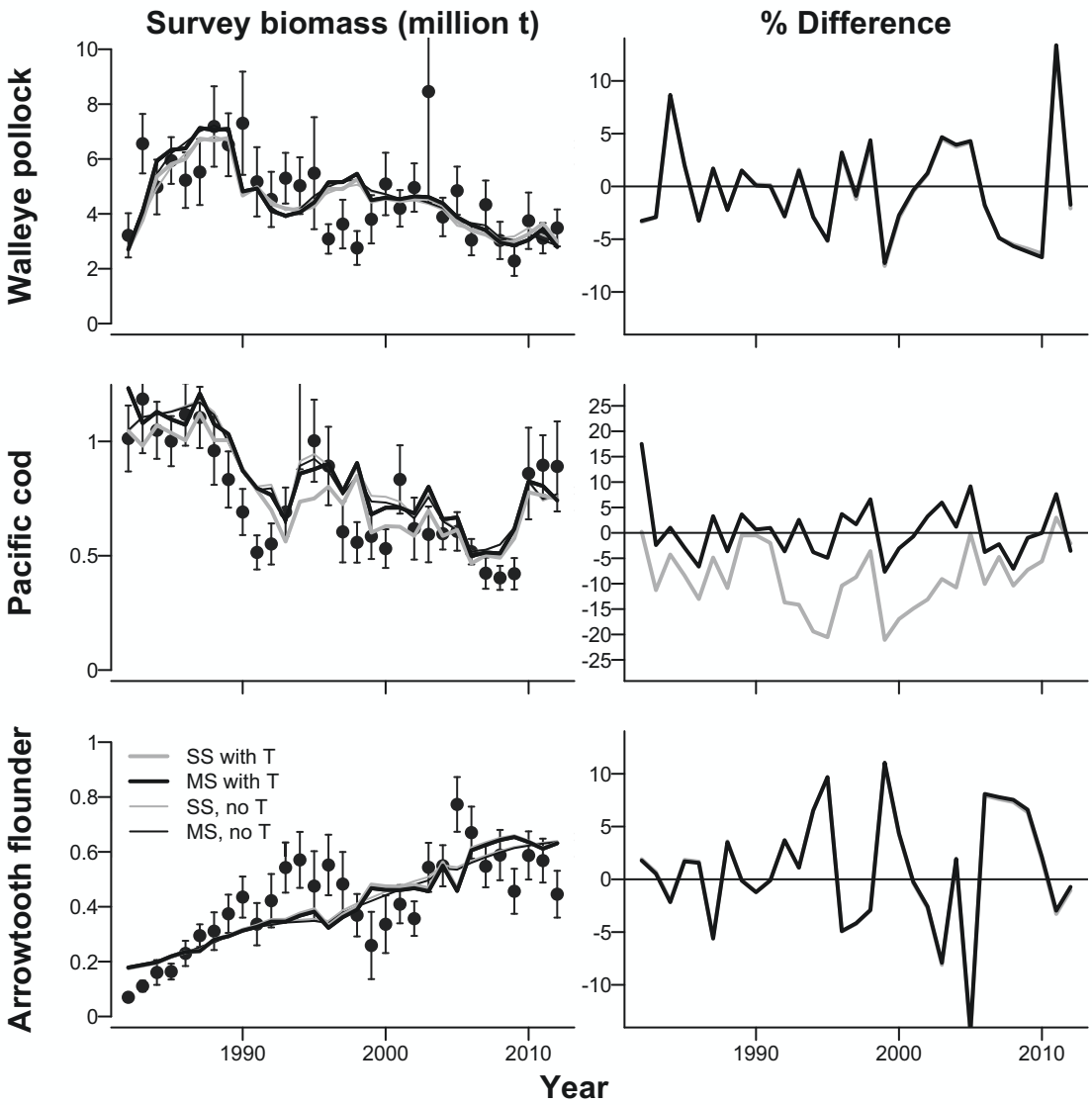

single-species
multi-species
Pollock
P. cod
Arrowtooth
Fishing mortality (F)
0.5
0.4
0.3
0.2
0.1
0
1980
1990
2000
2010
Year

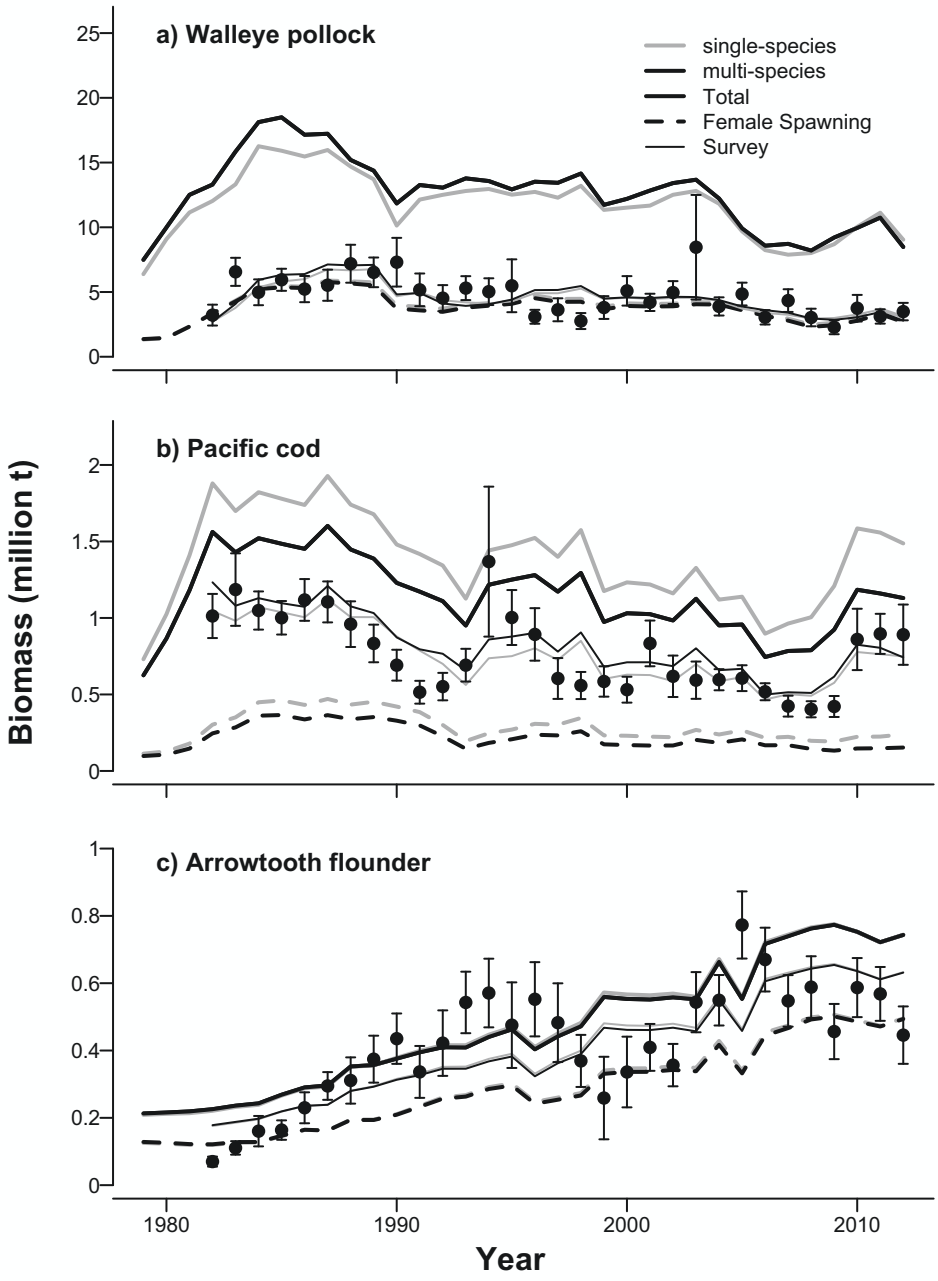

a) Walleye pollock
single-species
multi-species
300
200
100
0
b) Pacific cod
2
1.5
1
0.5
0
c) Arrowtooth flounder
0.8
0.6
0.4
0.2
0
1980
1985
1990
1995
2000
2005
2010
Recruitment (billions)
Year

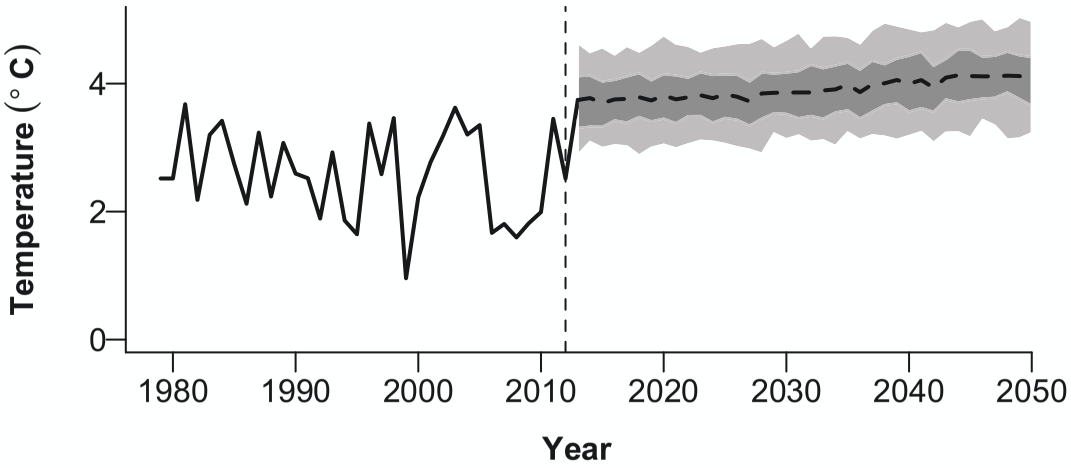

Walleye pollock
Pacific cod
Arrowtooth flounder
Proportion of total M2
1.0
0.8
0.6
0.4
0.2
0.0
1980
1985
1990
1995
2000
2005
2010
Year

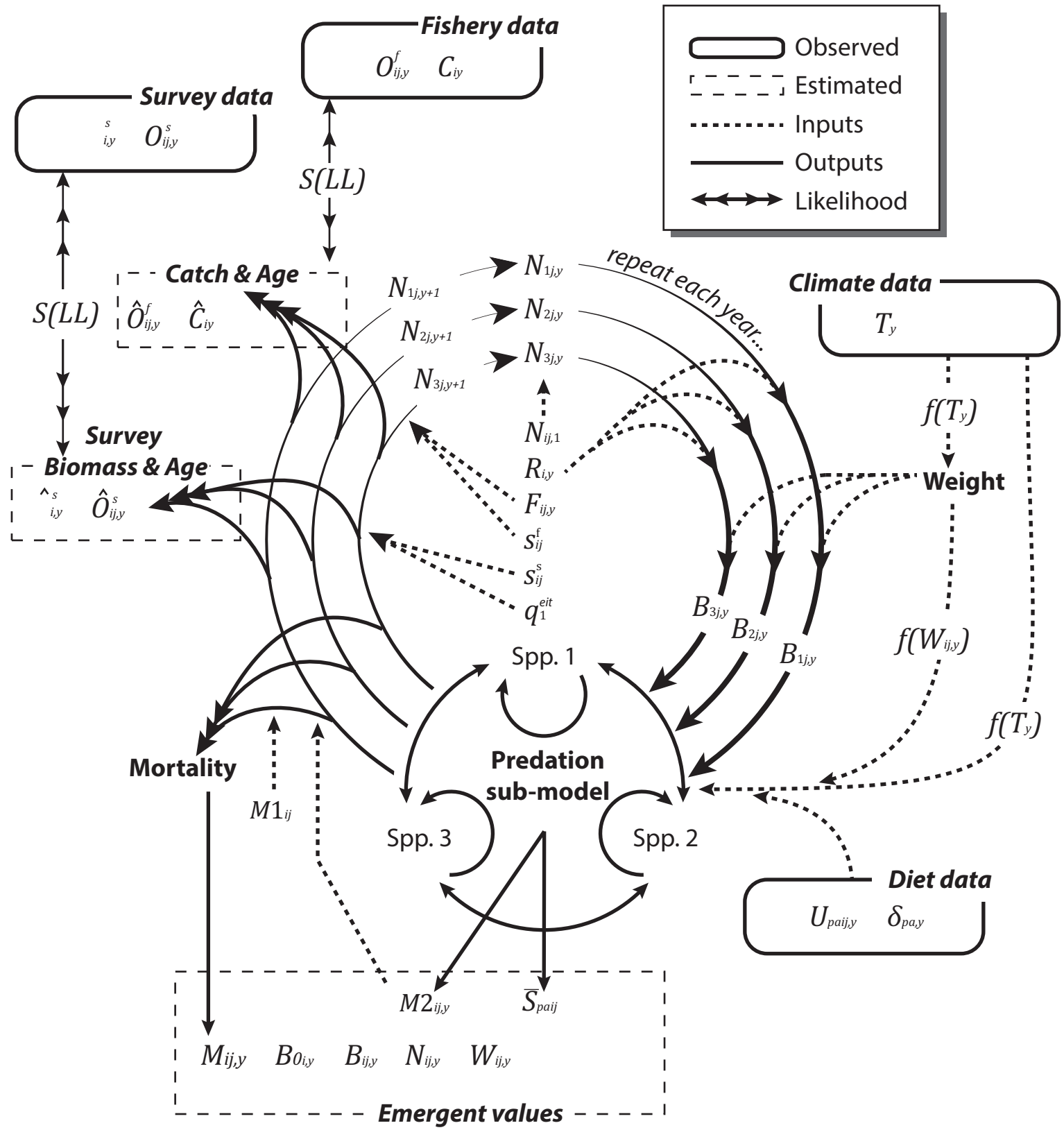

a)
Walleye pollock
Pacific cod
Arrowtooth founder
other prey
Predator ration (million t)
0
2
4
6
8
1980
1985
1990
1995
2000
2005
2010
b)
Prey consumed (thousand t)
0.001
0.01
0.1
1
10
100
1000
10000
c)
Pollock M2 (age 1)
0
0.5
1
1.5


a) Walleye pollock
b) Pacific cod
c) Arrowtooth flounder
single-species (i.e., M1)
multi-species, with T
multi-species, without T
Age 1 natural mortality (M1+M2)
Year
0
1
2
3
1980
1990
2000
2010


Survey biomass (million t)
% Difference
Walleye pollock
Pacific cod
Arrowtooth flounder
SS with T
MS with T
SS, no T
MS, no T
1990
2000
2010
Year


a) Walleye pollock
single-species
multi-species
Total
Female Spawning
Survey
b) Pacific cod
c) Arrowtooth flounder
Biomass (million t)
Year
0
5
10
15
20
25
0.5
1
1.5
2
0.2
0.4
0.6
0.8
1980
1990
2000
2010


Temperature (° C)
0
2
4
1980
1990
2000
2010
2020
2030
2040
2050
Year


Observed
Estimated
Inputs
Outputs
Likelihood
Fishery data
$O^f_{ij,y}$ C_{iy}
Survey data
$^s_{i,y}$ $O^s_{ij,y}$
$S(LL)$
Catch & Age
$\hat{O}^f_{ij,y}$ $\hat{C}_{iy}$
$S(LL)$
Survey
Biomass & Age
$\hat{}^s_{i,y}$ $\hat{O}^s_{ij,y}$
$N_{1j,y+1}$
$N_{2j,y+1}$
$N_{3j,y+1}$
$N_{1j,y}$
$N_{2j,y}$
$N_{3j,y}$
repeat each year...
$N_{ij,1}$
$R_{i,y}$
$F_{ij,y}$
s^f_{ij}
s^s_{ij}
q^{eit}_1
Climate data
T_y
$f(T_y)$
Weight
$B_{3j,y}$
$B_{2j,y}$
$B_{1j,y}$
$f(W_{ij,y})$
$f(T_y)$
Spp. 1
Spp. 3
Spp. 2
Predation
sub-model
Mortality
$M1_{ij}$
Diet data
$U_{paij,y}$ $\delta_{pa,y}$
$M2_{ij,y}$
$\overline{S}_{paij}$
$M_{ij,y}$ $B0_{i,y}$ $B_{ij,y}$ $N_{ij,y}$ $W_{ij,y}$
Emergent values